

COMPUTER-ASSISTED PROOFS OF CHAOS IN THE LONG SHORT-TERM
MEMORY MODEL

by

Duncan Tai Bennett

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As members of the Dissertation Committee, we certify that we have read the dissertation prepared by: Duncan Tai Bennett, titled: *Computer-Assisted Proofs of Chaos in the Long Short-Term Memory Model*.

and recommend that it be accepted as fulfilling the dissertation requirement for the Degree of Doctor of Philosophy.

Marek R Rychlik

Marek Rychlik

[Signature]
Kevin Lin (Jul 16, 2025 14:11 PDT)

Kevin Lin

Michael Chertkov

Misha Chertkov

Qiu-Dong Wang
Qiu-Dong Wang (Jul 15, 2025 11:05 PDT)

Quidong Wang

Date: 07/15/2025

Date: 07/16/2025

Date: 07/15/2025

Date: 07/15/2025

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I hereby certify that I have read this dissertation prepared under my direction and recommend that it be accepted as fulfilling the dissertation requirement.

Marek R Rychlik

Marek Rychlik
Dissertation Committee Chair
Mathematics

Date: 07/15/2025

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DEDICATION

Dedicated to my wife Samantha and my daughter Poppy.

Contents

List of Figures	7
ABSTRACT	8
CHAPTER 1 Introduction	9
1.1 Motivation	10
CHAPTER 2 Rank One Model for LSTM	12
2.1 The LSTM Model in Machine Learning	12
2.2 LSTM as a Dynamical System	14
2.3 Symptomatic Evidence of Chaos	17
2.3.1 Hénon-type Folding of Foliations	17
2.3.2 Positive Lyapunov Exponents and Period Doubling Cascade	18
2.3.3 Forced LSTM	21
CHAPTER 3 Absolutely Continuous Invariant Measures for the One Pa- rameter LSTM Family	24
3.1 Topological Horseshoes	28
3.1.1 Zgliczynski's Approach	29
3.1.2 Direct Approach	34
3.2 Continuations and Hitting the Horseshoe	36
3.2.1 Continuation of Horseshoe Points	36
3.2.2 Hitting the Horseshoe	39
3.3 One Dimensional Admissible Family	40
3.3.1 Parameter Transversality	40

CHAPTER 4 Numerical Methods and Analysis	43
4.1 Well Definedness of the family $\Psi_{a,b}$	44
4.2 Topological Horseshoe	46
4.3 Schwarzian Derivative of LSTM	48
4.4 Critical Orbit Lands in Horseshoe	49
4.5 Parameter Transversality	50
CHAPTER 5 A Conjecture for the Two Parameter Family $\Psi_{a,b}$	53
CHAPTER 6 Discussion	55
APPENDIX A Interval Analysis	57
A.1 The Interval Number System	57
A.2 Interval Functions and Sequences	58
A.3 Interval Newton's Method	60
A.4 Machine Interval Arithmetic	61
APPENDIX B Formulae for LSTM	64
APPENDIX C One Dimensional and Symbolic Dynamics	66
C.1 One Dimensional Dynamics	66
C.2 Symbolic Dynamics	66
APPENDIX D On Theory of Rank One Chaos	68
REFERENCES	70

List of Figures

2.1	One dimensional LSTM with horseshoe cover	16
2.2	Bifurcation diagram for parameter a	18
2.3	After a few iterations, we can see Hénon-type folding of the state space. Top Left: Initial foliation to be iterated by the LSTM map. Top Right: After 1 iteration. Bottom Left: After 3 iterations. Bottom Right: After 4 iterations.	19
2.4	Bifurcation diagram for parameter a	23

ABSTRACT

Long short-term memory (LSTM) is a type of recurrent neural network (RNN) that is used in sequence modeling which aims to mitigate the vanishing gradient problem. When the input is a constant zero sequence, the LSTM model can be viewed as a discrete dynamical system which can exhibit chaotic properties. In this article, we take a closer look at these chaotic properties which include period doubling cascades and positive Lyapunov exponents. We also present computer-assisted proofs of chaos in this model. For the one-dimensional LSTM family we start by proving the existence of a topological horseshoe with positive topological entropy using methods developed by [Zgliczynski \(1997\)](#) for multiple parameters. Then, using this horseshoe, we prove the existence of an absolutely continuous invariant measure with positive Lyapunov exponent for a positive measure set of parameters. Finally, we conjecture, for the two-dimensional family, the existence of an SRB measure for a positive measure set of parameters.

Chapter 1

Introduction

In this paper we introduce the long short-term memory model (LSTM) as a dynamical systems and prove, with computer assistance, that it is chaotic. We start by defining a two parameter two dimensional family of dynamical systems $\Psi_{a,b}$ (2.3) related to the Long Short-Term Memory model (2.1) from machine learning. When $b = 0$, this family can be reduced to a one dimensional family Ψ_a (4.4). We first show that, for certain parameters, Ψ_a contains a topological horseshoe with positive topological entropy. Using this fact, we prove that the family Ψ_a contains uncountably many Misiurewicz maps. In addition, subject to some technical requirements, we prove that for a set of parameters with positive Lebesgue measure Ψ_a admits absolutely continuous invariant measures with positive Lyapunov exponent. We then conjecture that for b sufficiently small, $\Psi_{a,b}$ admits an SRB measure. For more on SRB measures see chapter 5.

Theorem 1.0.1 (Topological Horseshoe in One Dimension). Let Ψ_a be as in (4.4) with parameters as in 4.1. There exists a compact subset $N \subset I$ such that $f_a|_{\text{Inv}(N, f_a)}$ is conjugate to a shift of finite type with positive topological entropy.

Theorem 1.0.2 (Absolutely Continuous Invariant Measures in One Dimension). Let Ψ_a be as in (4.4) with parameters as in 4.1. There exists a positive measure set of parameters $\Delta \subset [a_1, a_2]$ such that for $a^* \in \Delta$, Ψ_{a^*} has an absolutely continuous invariant probability measure.

Conjecture 1.0.1 (SRB Measures in Two Dimensions). Let $\Psi_{a,b}$ be as in (2.3) with parameters as in 4.1 and let a^* be as in theorem 1.0.2. Then for sufficiently small b , $\Psi_{a^*,b}$ admits an SRB measure.

1.1 Motivation

In this section, we provide some context for the above results and why practitioners of machine learning should care about such theory. We start by outlining the general setup in supervised learning. Then we discuss some intuition behind the LSTM model and present a high level idea of the significance of our analysis. A more in depth analysis of these results is left to the section 6.

In machine learning, specifically supervised learning, the goal is to learn/describe/generalize a relationship between input and output data. This problem setup requires the following: training data $(\mathbf{X}, \mathbf{y})$, a family of functions $\{f_\theta\}_{\theta \in \Theta}$, and a loss function $\mathcal{L}(\mathbf{X}, \mathbf{y}, f_\theta)$. We then wish to solve or approximate solutions to the optimization problem $\min_{\theta \in \Theta} \mathcal{L}(\mathbf{X}, \mathbf{y}, f_\theta)$. In practice, we must make choices for each one of these items. For example, we can transform or project data, we can penalize different loss functions or we can choose different approximation techniques for solving the minimization problem. In this paper, we are concerned with the choice of the family $\{f_\theta\}_{\theta \in \Theta}$ which can vary from linear, polynomial and neural networks with a variety of architectures. Much work has been put into designing different architectures for these functions using heuristics and empirical evidence for different use cases but there is less emphasis on their inherent properties. We wish to address some of these inherent properties for the Long Short-Term Memory (LSTM) model.

The LSTM model was developed in [Hochreiter and Schmidhuber \(1997\)](#) in an attempt to remedy the vanishing gradient problem. This problem occurs in recurrent neural networks or deep networks due to backpropagation and underflow. To illustrate this, suppose $f(x_t, h_{t-1}) = h_t$ is a recurrent neural network. The impact x_1 has on h_N for N large is the derivative $\frac{\partial h_N}{\partial x_1} = \frac{\partial h_1}{\partial x_1} \prod_{i=2}^N \frac{dh_i}{dh_{i-1}}$. In particular, each factor will contain the weights of f as a factor. For example, if $f(x, h) = \sigma(ax + bh)$ the each factor contains b which will cause the gradient to vanish (or explode) if $|b| < 1$ (or $|b| > 1$).

If underflow occurs in this large product, then the derivative cannot be recovered for future values of h_n , $n > N$. An analogous problem occurs with overflow which we call the exploding gradient problem.

The analysis above can be interpreted roughly as “recurrent neural networks don’t have long term memory”. At some point, h_t does not depend on x_s for $t - s$ large. This is clearly an issue when there exists long range dependencies in the data. This problem is addressed in the LSTM model by adding two hidden activations: the hidden state and cell state, which function as a short term and long term memory respectively. To see an in depth description of this see [Hochreiter and Schmidhuber \(1997\)](#).

In our analysis, we address dynamics of the hidden and cell state of the LSTM model for fixed parameters. In particular, if we set the input sequence $x_t = 0$ then the LSTM models is $\text{LSTM}(h_t, c_t) = (h_{t+1}, c_{t+1})$, a discrete dynamical systems. Analyzing the dynamics of this systems is to ask “if no new information is given to the model, how are the short and long term memories impacted?” We may posit that the dynamics of these states should be simple or predictable since no new information is provided to update them but this is not the case. As we will see, even with zero input, the above dynamics can be chaotic.

Chapter 2

Rank One Model for LSTM

In this section, we present the long short-term memory model and our interpretation of it as a dynamical system. In section 2.1, we start by introducing the formulation from the perspective of machine learning and then make the necessary modifications and simplifications to view it as a discrete dynamical system. In section 2.2, we present the final formulation of the LSTM family that we will use for the rest of the paper. Lastly, in section 2.3 we present some symptomatic numerical evidence of chaos, namely period doubling cascades, positive Lyapunov exponents and Hénon-type folding of foliations. These are found with naive (non-rigorous) numerics in contrast to later numerical simulations we will present.

2.1 The LSTM Model in Machine Learning

In this section, we introduce the LSTM model as it is used in machine learning. The specific architecture of the model is designed to mitigate the vanishing gradient problem that depends on two states, the hidden state and the cell state, which metaphorically represent short and long term memories.

These long term memories will be resistant to the vanishing gradient problem in the following way. In reference to equations (2.1), if we compute $\frac{dc_t}{dc_{t-1}} = f_t + i_t \sigma'(W_c x_t + U_c h_{t-1} + b_c) U_c \frac{dh_{t-1}}{dc_{t-1}}$ we see that there is no shared factor among the terms. This shared factor, as we saw in the introduction discussion, was a systematic cause of the vanishing/exploding of the gradient. Instead, the persistence of the cell state only depends on the activations. The above is only meant to suggest that this remedies the vanishing gradient problem. For a more in depth discussion see [Hochreiter and Schmidhuber \(1997\)](#).

Now we present the formal definition of the LSTM model. Let $x_t \in \mathbb{R}^d$ be an input sequence and let $W_\alpha \in \mathbb{R}^{n \times d}$, $U_\alpha \in \mathbb{R}^{n \times n}$ and $b_\alpha \in \mathbb{R}^n$ be weights for $\alpha \in \{f, i, c, o\}$. These indices refer to the forget gate, input gate, cell state and output gate respectively. Then the sequences $h_t, c_t \in \mathbb{R}^n$, called the hidden state and cell state respectively, are determined by the LSTM model,

$$\begin{aligned}
 f_t &= \sigma_g(W_f x_t + U_f h_{t-1} + b_f) \\
 i_t &= \sigma_g(W_i x_t + U_i h_{t-1} + b_i) \\
 o_t &= \sigma_g(W_o x_t + U_o h_{t-1} + b_o) \\
 \tilde{c}_t &= \sigma_c(W_c x_t + U_c h_{t-1} + b_c) \\
 c_t &= f_t \odot c_{t-1} + i_t \cdot \tilde{c}_t \\
 h_t &= o_t \odot \sigma_h(c_t).
 \end{aligned} \tag{2.1}$$

A typical choice for the activation functions $\sigma_g, \sigma_c, \sigma_h$ are the sigmoid $\sigma(x) = (1 + e^{-x})^{-1}$ for σ_g and $\tanh$ for σ_c and σ_h . Here, $\odot$ denotes the Hadamard product or component-wise product.

In our study, we set $b_\alpha = 0$ for $\alpha \in \{f, i, c, o\}$ and $x_t = 0$ for all t . This reduces the model to

$$\begin{aligned}
 f_t &= \sigma_g(U_f h_{t-1}) \\
 i_t &= \sigma_g(U_i h_{t-1}) \\
 o_t &= \sigma_g(U_o h_{t-1}) \\
 \tilde{c}_t &= \sigma_c(U_c h_{t-1}) \\
 c_t &= f_t \odot c_{t-1} + i_t \cdot \tilde{c}_t \\
 h_t &= o_t \odot \sigma_h(c_t).
 \end{aligned}$$

This can be rewritten as

$$\begin{bmatrix} h_t \\ c_t \end{bmatrix} = \text{LSTM} \left(\begin{bmatrix} h_{t-1} \\ c_{t-1} \end{bmatrix} \right) = \begin{bmatrix} \sigma(U_o h_{t-1}) \tanh[\sigma(U_f h_{t-1}) c_{t-1} + \sigma(U_i h_{t-1}) \tanh(U_c h_{t-1})] \\ \sigma(U_f h_{t-1}) c_{t-1} + \sigma(U_i h_{t-1}) \tanh(U_c h_{t-1}) \end{bmatrix} \quad (2.2)$$

For the rest of the paper, we make the following notational substitutions

$$x = h$$

$$y = c$$

$$U_f = b$$

$$U_o = a$$

$$U_i = k_1$$

$$U_c = k_2.$$

We will also consider the case where our dimensions are as small as possible. That is $d = n = 1$ so that the hidden state x and the cell state y are just real numbers (as well as the weights).

2.2 LSTM as a Dynamical System

The LSTM model as presented at the end of 2.1, can be written as

$$\begin{aligned} \text{LSTM}_{a,b}(x, y) &= \begin{bmatrix} \sigma(bx) \tanh((G(x, y, a))) \\ G(x, y, a) \end{bmatrix} \\ G(x, y, a) &= \sigma(ax)y + \sigma(k_1 x) \tanh(k_2 x) \end{aligned}$$

where k_1, k_2 are fixed. In our case, we will use $k_1 = -6.105$ and $k_2 = -12.21$. We can see that when $b = 0$ the dynamics occur on the graph of $x = \frac{1}{2} \tanh(y)$. We apply a shear map $\Phi(x, y) = (x + \frac{1}{2} \tanh(y), y)$ to $\Psi_{a,b}$ so that these dynamics occur on the y -axis. That is,

$$\begin{aligned}
& \Phi^{-1} \circ \text{LSTM}_{a,b} \circ \Phi(x, y) \\
&= \Phi^{-1} \circ \text{LSTM}_{a,b} \begin{bmatrix} x + \frac{1}{2} \tanh(y) \\ y \end{bmatrix} \\
&= \Phi^{-1} \circ \begin{bmatrix} \sigma(b(x + \frac{1}{2} \tanh(y)))t((G(x + \frac{1}{2} \tanh(y), y, a)) \\ G(x + \frac{1}{2} \tanh(y), y, a) \end{bmatrix} \\
&= \begin{bmatrix} \sigma(b(x + \frac{1}{2} \tanh(y))) \tanh((G(x + \frac{1}{2} \tanh(y), y, a)) - \frac{1}{2} \tanh(G(x + \frac{1}{2} \tanh(y), y, a)) \\ G(x + \frac{1}{2} \tanh(y), y, a) \end{bmatrix} \\
&= \begin{bmatrix} (\sigma(b(x + \frac{1}{2} \tanh(y))) - \frac{1}{2}) \tanh((G(x + \frac{1}{2} \tanh(y), y, a)) \\ G(x + \frac{1}{2} \tanh(y), y, a) \end{bmatrix}
\end{aligned}$$

We will refer to this dynamical system as the two parameters LSTM family.

Definition 2.2.1. We define the *two parameter LSTM family* as $\Psi_{a,b} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$$\begin{aligned}
\Psi_{a,b}(x, y) &= \begin{bmatrix} (\sigma(b(x + \frac{1}{2} \tanh(y))) - \frac{1}{2}) \tanh(G(x + \frac{1}{2} \tanh(y), y, a)) \\ G(x + \frac{1}{2} \tanh(y), y, a) \end{bmatrix} \quad (2.3) \\
G(x, y, a) &= \sigma(ax)y + \sigma(k_1x) \tanh(k_2x).
\end{aligned}$$

We will commonly restrict the domain to a trapping region, which we will denote $D \times I \subset \mathbb{R}^2$. We may also use the alternative notation $\Psi(x, y; a, b) = \Psi_{a,b}(x, y)$.

When $b = 0$, $\Psi_{a,0}(x, y) \in \{0\} \times \mathbb{R}$ and thus we can reduce the model to a one dimensional map. See Figure 2.1 for an example graph.

Definition 2.2.2. We define the *one parameter LSTM family* as $\Psi_a : \mathbb{R} \rightarrow \mathbb{R}$

$$\begin{aligned}
\Psi_a(y) &= G\left(\frac{1}{2} \tanh(y), y, a\right) \quad (2.4) \\
G(x, y, a) &= \sigma(ax)y + \sigma(k_1x) \tanh(k_2x).
\end{aligned}$$

We will commonly restrict the domain to a trapping region, which we will denote $I \subset \mathbb{R}$. We may also use the alternative notation $\Psi(y; a) = \Psi_a(y)$.

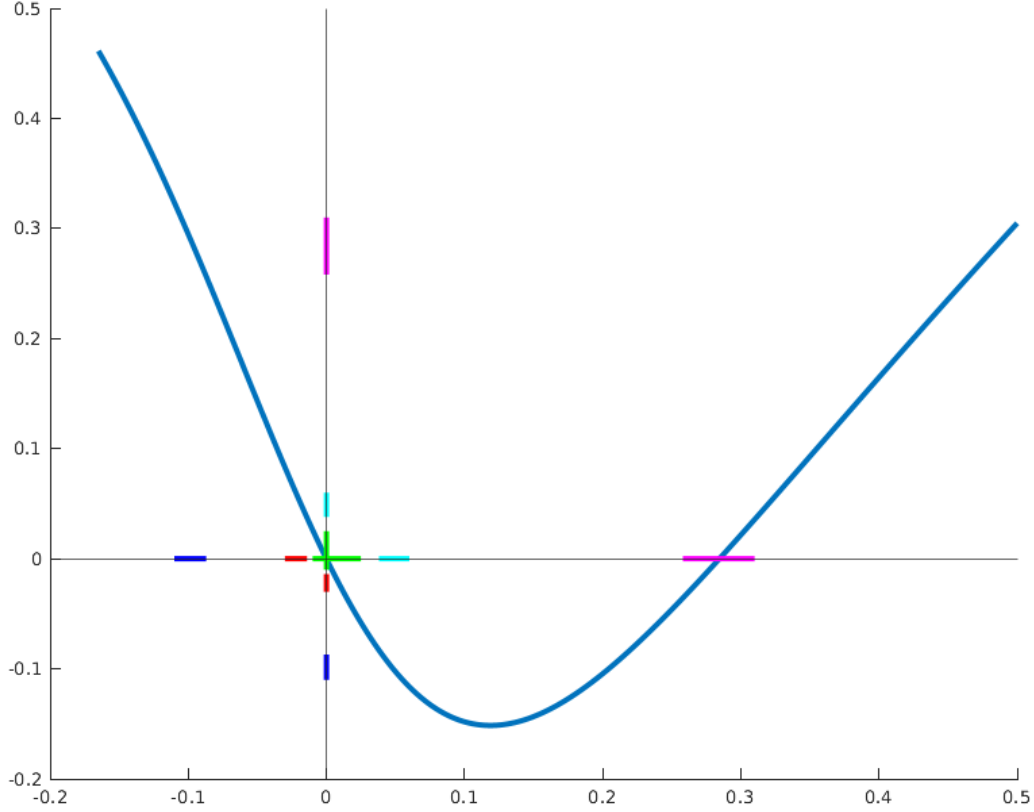


Figure 2.1: Plot of the function Ψ_a for parameter $a = 31$ where $k_1 = -6.105$, $k_2 = -12.21$ and $b = 0$. In addition, the horseshoe cover from lemma 4.2.1 and lemma 4.2.2 are shown as the colored intervals.

Lemma 2.2.1. Let parameters be as in 4.2. The two parameter LSTM family is well-defined and smooth on $D \times I$ for all parameters $(a, b) \in [a_1, a_2] \times [b_1, b_2]$.

Proof.

Well-definedness is to show that $\Psi_{a,b}(D \times I) \subset D \times I$ which is proven by computer-assistance in lemma 4.1.1. Smoothness follows from the fact that we take a compositions of σ , $\tanh$ and affine maps which are all smooth. $\square$

Lemma 2.2.2. Let parameters be as in 4.2. The one parameter LSTM family is well-defined and smooth on I for all parameters $a \in [a_1, a_2]$.

Proof.

This follows from lemma 2.2.1. □

It's important to note going forward that the above two results imply the existence of an invariant measure for $\Psi_{a,b}$ and Ψ_a for all relevant parameters. This follows from the fact that continuous dynamical systems on compact domains have invariant measures due to the compactness of the space of probability measures. This is left as an exercise or can be found in [Boyarsky and Gora \(2012\)](#).

2.3 Symptomatic Evidence of Chaos

In this section we present symptomatic, numerical evidence of chaos in the LSTM model. Under certain parameters, we see Hénon-type folding of the phase space, period doubling cascades and (non-rigorously computed) positive Lyapunov exponents. This evidence for chaos has been addressed in papers such as [Ribeiro et al. \(2020\)](#) but has not been fully investigated as we will do in the following sections. We conclude this section with a brief discussion of a forced LSTM. That is, what happens when the input sequence of the LSTM model is not constant zero.

2.3.1 Hénon-type Folding of Foliations

To visualize the effects $\text{LSTM}_{a,b}$ has on the state space $\mathbb{R}^2$ we track the mapping of 1 dimensional foliations. In particular, we track the evolution of lines $y = \text{constant}$ which is done by plotting the grid

$$A = \{(x, y) \in \mathbb{R}^2 : x = n \cdot \Delta x + m \cdot \Delta y, (n, m) \in \mathbb{Z}^2 \cap [-5, 5]^2\} \quad (2.5)$$

with $\Delta x = 0.01$ and $\Delta y = 0.1$. We color the lines $y = \text{constant}$ each different colors to better track the evolution of that foliation. In Figure 2.3 we can see the first few iterations of the

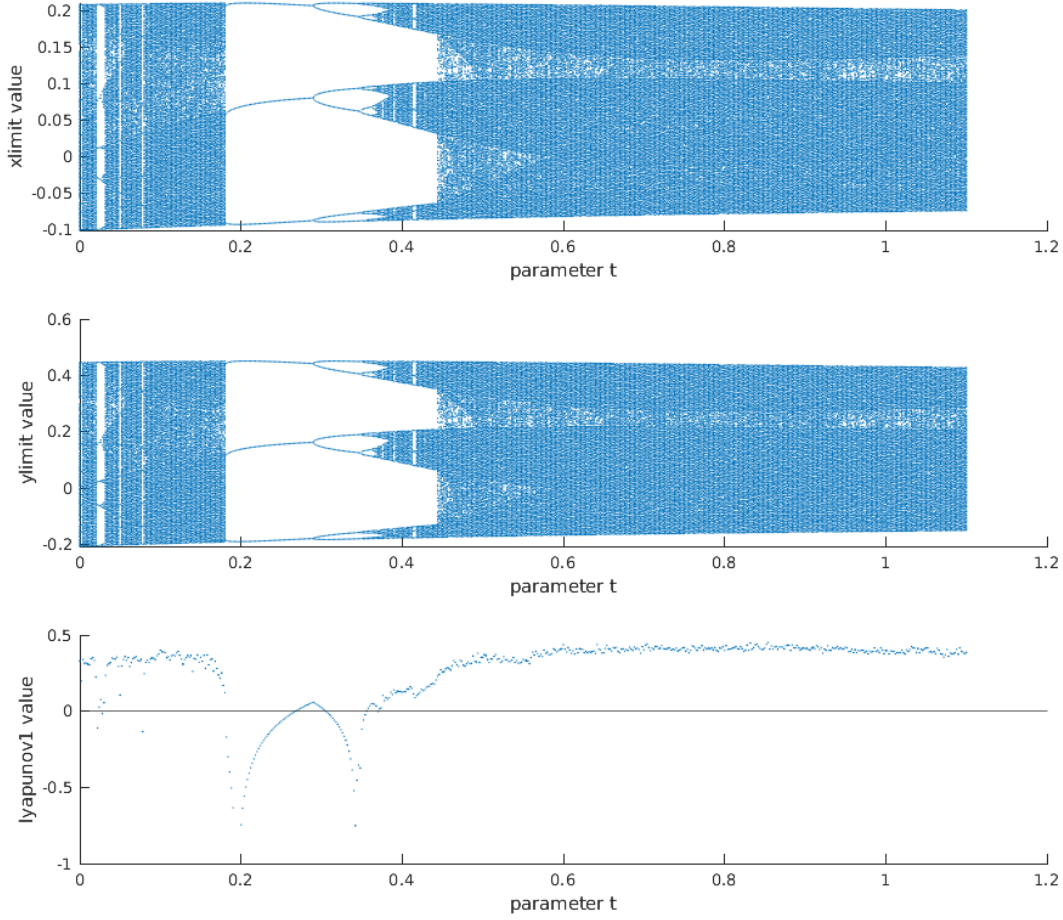


Figure 2.2: Bifurcation diagram for parameter $a = 30t$ where $k_1 = -6.105$, $k_2 = -12.21$ and $b = 0$. The orbits shown are for the point $(0.5, 0.5)$.

$\text{LSTM}_{30,0}$ map on a foliation. This illustrates the Hénon-type folding behavior that is characteristic of some 2 dimensional chaotic mappings. The parameter $a = 30$ was chosen using results from the experiment in the following section. In particular, one can see the potential chaotic region of parameter space in Figure 2.2.

2.3.2 Positive Lyapunov Exponents and Period Doubling Cascade

In this section, we present our computation of positive Lyapunov exponents and a bifurcation diagram showing the period doubling cascade of $\text{LSTM}_{a,b}$. A period doubling cascade

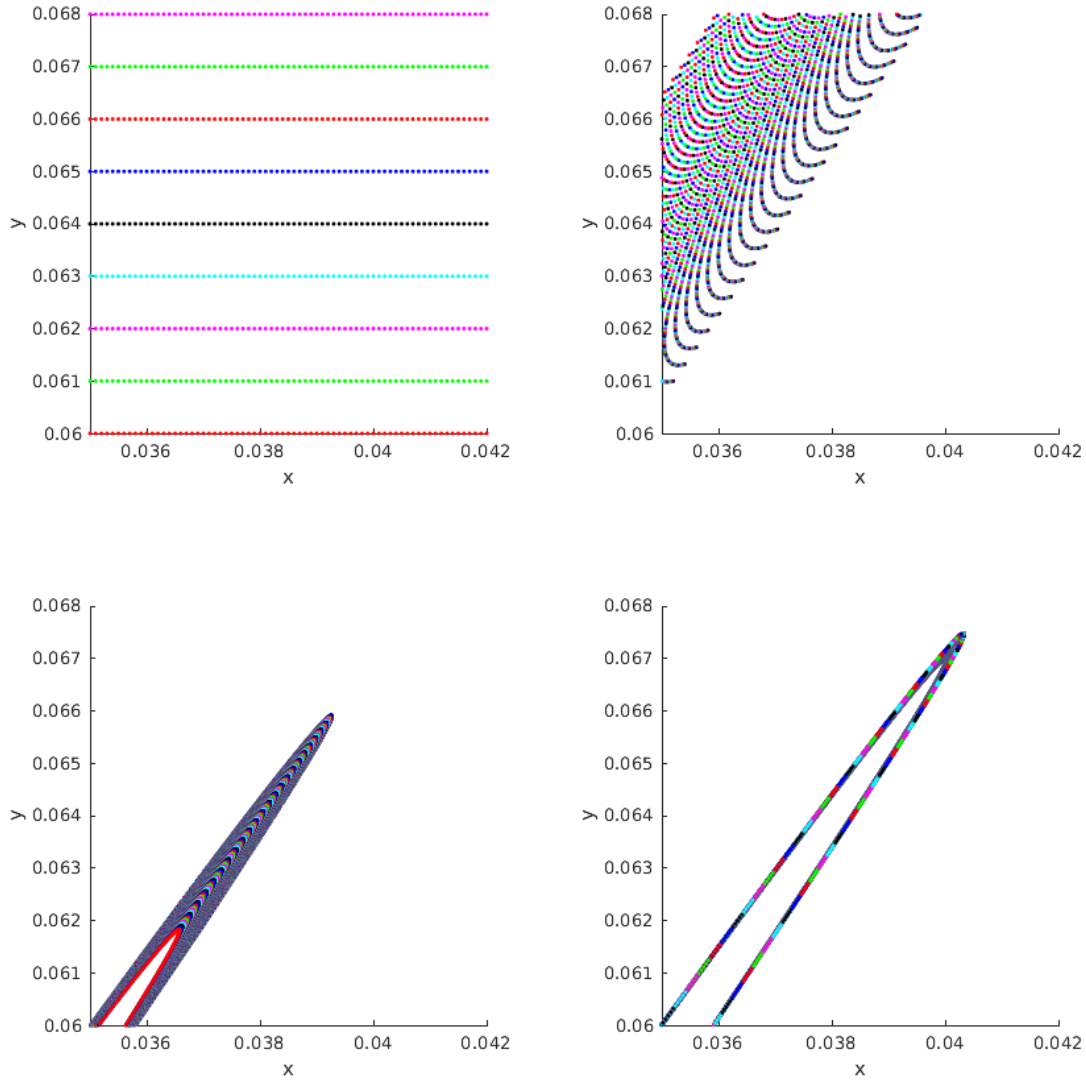


Figure 2.3: After a few iterations, we can see Hénon-type folding of the state space. **Top Left:** Initial foliation to be iterated by the LSTM map. **Top Right:** After 1 iteration. **Bottom Left:** After 3 iterations. **Bottom Right:** After 4 iterations.

is an (infinite) sequence of period doubling bifurcations and is commonly a route to chaos. We first present the set up of our experiment and then present the results.

In this experiment, we track the limit behavior of our model for various values of $a = 30t$ and $t \in [0, 1.2]$ as well as approximate the corresponding Lyapunov exponents for $(x_0, y_0) = (1/2, 1/2)$. We capture the limit behavior by looking at the ω -limit set $\omega((1/2, 1/2), \text{LSTM}_{a,0})$ which is approximated by a finite set. In addition, the Lyapunov exponent is approximated using this finite set, as we will see below.

To calculate an approximate ω -limit set, we burnin in the dynamical system for N iterations and then take M further iterations. That is, our approximate ω -limit set W_{x_0} is

$$W_{x_0}(N, M) = \{x : x = f^n(x_0), n \in \mathbb{Z} \cap [N, N + M]\} \quad (2.6)$$

To approximate the Lyapunov exponent of x_0 , first define the operator

$$T_{N,M}(x_0) = \prod_{i=N+1}^{N+M} D_x(f)(f^{i-1}(x_0)).$$

Then, we may define the *first Lyapunov exponent* as

$$\lambda_1 = \lim_{M \rightarrow \infty} \frac{1}{M} \log (\|T_{0,M}(x_0)\|)$$

where $\|\cdot\|$ is the spectral norm. This limit is guaranteed to exist μ almost everywhere by Oseledec's multiplicative ergodic theorem under certain conditions, where μ is an invariant measure of f .

To approximate the first Lyapunov exponent, we use a method of [Eckmann and Ruelle \(1985\)](#). This method uses the fact that $T_{0,M}$ can be written as

$$T_{0,M}(x) = Q_M R_M R_{M-1} \cdots R_1$$

where each R_i is upper triangular and Q is orthogonal. In addition, the largest diagonal element r_M of $R = R_M \cdots R_1$ is such that

$$\lim_{M \rightarrow \infty} \log(r_M) = \lambda_1$$

It should be noted that we may replace $T_{0,M}$ with $T_{N,M}$ in the above computations (as long as these factors are nonzero). More information on this method can be found in [Eckmann and Ruelle \(1985\)](#).

To produce our bifurcation diagram and Lyapunov exponents we partition the parameter space $[1/2, 3]$ with $\Delta a = 0.005$. Then for each a , we approximate the ω -limit set $\omega((1/2, 1/2), \text{LSTM}_{a,0})$ by

$$W_{(1/2, 1/2)}(N, M) = \{(x, y) : (x, y) = \text{LSTM}_{a,0}^n(1/2, 1/2), n \in \mathbb{Z} \cap [N, N + M]\}$$

where the burnin is $N = 3000$ and the number of points in the approximated ω -limit set is $M = 2000$. The Lyapunov exponent of $\text{LSTM}_{a,0}$ for $(x_0, y_0) = (1/2, 1/2)$ is calculated using the spectrum of the approximate $T = \prod_{i=N+1}^{N+M} D_x(\text{LSTM}_{a,0})(\text{LSTM}_{a,0}^{i-1}(1/2, 1/2))$ with the same values of N and M .

2.3.3 Forced LSTM

We conclude our section on numerical experiments with an experiment for a forced LSTM model. If we compare the model from equation (2.2) with

$$\begin{aligned} \begin{bmatrix} h_t \\ c_t \end{bmatrix} &= \text{LSTM} \left(\begin{bmatrix} h_{t-1} \\ c_{t-1} \\ x_t \end{bmatrix} \right) \\ &= \begin{bmatrix} \sigma(U_o h_{t-1} + W_o x_t) \tanh[\sigma(U_f h_{t-1} + W_f x_t) c_{t-1} + \sigma(U_i h_{t-1} + W_i x_t) \tanh(U_c h_{t-1} + W_c x_t)] \\ \sigma(U_f h_{t-1} + W_f x_t) c_{t-1} + \sigma(U_i h_{t-1} + W_i x_t) \tanh(U_c h_{t-1} + W_c x_t) \end{bmatrix} \end{aligned} \quad (2.7)$$

we can metaphorically think of x_t as a forcing of the unforced dynamics in equation (2.2) resulting in the forced dynamics of equation (2.7). We can then ask how does this forcing affect the chaotic behavior of the LSTM (i.e. how is the Lyapunov exponent affected)?

As an example, we consider the case where x_t is determined by $u_{t+1} = g(u_t) = u_t + (\sqrt{3} - 1) \bmod 1$ and $x_t = \sin(2\pi u_t)$. The seed of this pseudo-random sequence will be $u_1 = 0.5$. We let $[U_f, U_i, U_c, U_o] = [10, 0, 0, 0]$ and $[W_f, W_i, W_c, W_o] = [30, -6.105, -12.21, 0]$ which are the same weights as in Figure 2.2.

To be a little more explicit, we consider the mapping on $\mathbb{R}^2 \times S^1$ defined by

$$(h, c, u) \mapsto (\text{LSTM}(h, c, \sin(2\pi u)), g(u)). \quad (2.8)$$

In addition, the Jacobian of this map will have eigenvectors that strictly lie in the (h, c) plane. We can then compute Lyapunov exponents (corresponding to vectors in the $(h, c,)$ plane) for this forced system as well. The diagram 2.4 is a bifurcation diagram which also shows the evolution of the first Lyapunov exponent.

We note that the positivity of the Lyapunov exponent persists with this forcing. However, this is not a complete examination of how input sequences affect the Lyapunov exponent. This example illustrates that the simplifications in the system (2.2) and its behavior is somewhat representative of the more general model.

In the bifurcation diagram 2.4 we see that the period double cascades have been replaced by bands that we hypothesize come from the rotation g . From the viewpoint of (2.8), we hypothesize that there exists an invariant circle in $\mathbb{R}^2 \times S^1$ that is observed in the above dynamics. This invariant circle would be approximately contained in $[-0.1, 0.2] \times [-0.2, 0.4] \times S^1$. This invariant circle may be a normally hyperbolic manifold (see Fenichel and Moser (1971)). In addition, there are intermittent stable orbits that are not obscured by the rotation. We currently have no explanation for this interesting behavior.

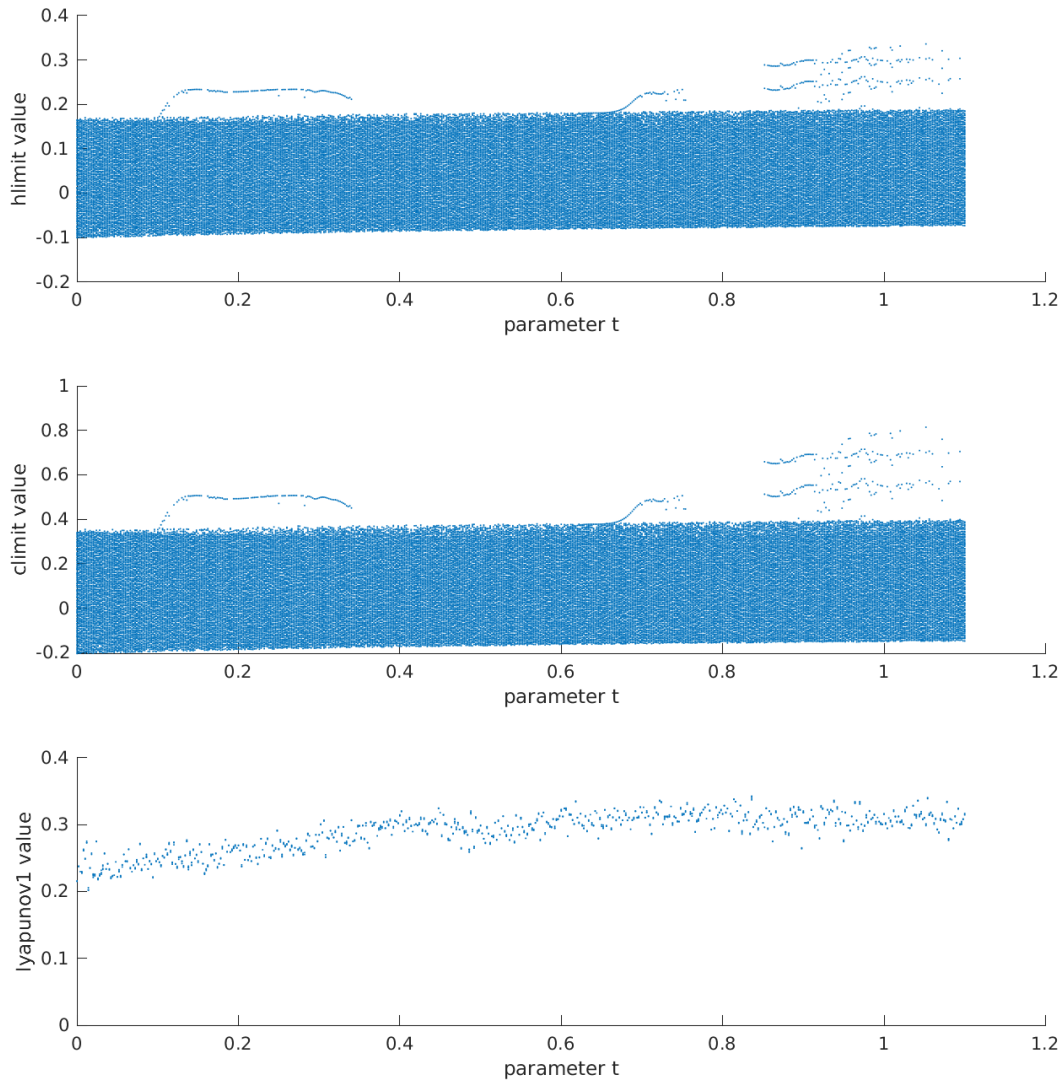


Figure 2.4: Bifurcation diagram for parameter $a = 30t$ where $k_1 = -6.105$, $k_2 = -12.21$ and $b = 0$. The orbits shown are for the point $(0.5, 0.5)$. The forcing parameters are $[U_f, U_i, U_c, U_o] = [10, 0, 0, 0]$.

Chapter 3

Absolutely Continuous Invariant Measures for the One Parameter LSTM Family

In this section, we present all theoretical findings required to show the existence of absolutely continuous invariant measures for the one dimensional LSTM family for a nonzero measure set of parameters. In section 3.1 we prove the existence of a topological horseshoe in the one dimensional family Ψ_a . Then in section 3.2 we prove some technical results that imply that, for certain parameters, the critical orbit lands in the aforementioned topological horseshoe. This fact is then used to show the existence of parameters a^* such that Ψ_{a^*} is a Misiurewicz map. In section 3.3 we verify a parameter transversality condition, which shows Ψ_a is an admissible family as defined in Wang and Young (2006). Such families contain a positive measure set of parameters such that the corresponding maps admit absolutely continuous invariant measures with respect to the Lebesgue measure. In each section, we will prove each theorem that can be proven directly and reference chapter 4 for any computer-assisted lemmas that are needed.

Our goal for this chapter will be to show that the Ψ family is an admissible family since such families admit the desired absolutely continuous invariant measures as in theorem D.0.2. More information on the rank-one theory can be found in appendix D.

We start with a brief presentation of some general results from the rank-one theory that we will leverage throughout this chapter. We will not need these results in their fullest generality so we will give the special cases we will use. The rank-one theory of Wang and Young (2006), Wang and Young (2008) depends on the construction of a family of maps which contain Misiurewicz maps. Such maps are defined below in their most generality, as well as some sufficient (but not necessary) conditions to show a map is a Misiurewicz map.

Definition 3.0.1 (Misiurewicz Map). Let $f \in C^2(I, I)$ where I is a closed interval or S^1 . Let C denote the set of critical points of f and $C_\delta = \{x \in I : \text{dist}(x, C) < \delta\}$ for some $\delta > 0$. We call f a *Misiurewicz map* if there exists some $\delta_0 > 0$ such that

- (a) Outside of C_{δ_0} : there exists $\lambda_0 > 1$, $M_0 \in \mathbb{N}$ and $0 < c_0 \leq 1$ such that
 - (i) for all $n \geq M_0$, if $x, f(x), \dots, f^{n-1}(x) \notin C_{\delta_0}$, then $|(f^n)'(x)| \geq e^{\lambda_0 n}$;
 - (ii) if $x, f(x), \dots, f^{n-1}(x) \notin C_{\delta_0}$ and $f^n(x) \in C_{\delta_0}$, any n , then $|(f^n)'(x)| \geq c_0 e^{\lambda_0 n}$.
- (b) Inside of C_{δ_0} :
 - (i) $f''(x) \neq 0$ for all $x \in C_{\delta_0}$;
 - (ii) for all $\hat{x} \in C$ and $n > 0$, $d(f^n(\hat{x}), C) \geq \delta_0$;
 - (iii) for all $x \in C_{\delta_0} \setminus C$, there exists $p_0(x) > 0$ such that $f^j(x) \notin C_{\delta_0}$ for all $j < p_0(x)$ and $|(f^{p_0(x)})'(x)| \geq c_0^{-1} e^{\frac{1}{3} \lambda_0 p_0(x)}$.

We denote the set of all Misiurewicz maps (on a given domain) by $\mathcal{M}$.

The salient feature of Misiurewicz maps are that they admit absolutely continuous invariant measures and yet they are nonuniformly expanding maps. This is due to the balance between expansion and contraction that allows for absolute continuity of the invariant measure. To illustrate this point, we consider a few examples where absolutely continuous invariant measures do not exist. In the case of a map with negative Schwarzian derivative, it can be shown that attractive periodic points must have a critical point in their basin of attraction. The existence of an attractive periodic point is problematic for an acim since attractive invariant sets will necessarily have zero Lebesgue measure. Thus, controlling the critical orbit, as well as contraction associated with attractive points, is key for the existence of acims in the nonuniformly expanding case. Without this negative Schwarzian condition, recurrence of the critical point can imply that it is a periodic attractor, or that the attractor is a Cantor set equal to the limit set of the critical point. For more information see [Melo and Strien \(1993\)](#).

This most general definition is hard to verify for a given family of maps. However, the family of maps Misiurewicz originally studied is slightly less general but the conditions are easy to verify. For this, we'll need the definition of the Schwarzian derivative.

Definition 3.0.2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a C^3 map such that $f' \neq 0$. The *Schwarzian derivative* is defined as

$$Sf(x) = \frac{f'''(x)}{f'(x)} - \frac{3}{2} \left(\frac{f''(x)}{f'(x)} \right)^2$$

Lemma 3.0.1. Suppose $f \in C^3(I, I)$ and let $C(f) = \{x \in I : f'(x) = 0\}$ be the critical set of f . If f satisfies

- (a) $Sf(x) < 0$ for all $x \in I \setminus C(f)$
- (b) $f''(\hat{x}) \neq 0$ for all $\hat{x} \in C(f)$
- (c) If $f^n(x) = x$ then $|(f^n)'(x)| > 1$
- (d) for all $\hat{x} \in C(f)$, $\inf(f^n(\hat{x}), C(f)) > 0$,

then $f \in \mathcal{M}$.

All conditions except for (c) above are simple to verify but by using Singer's theorem we can reduced the number of orbits we have to check to a finite number.

Theorem 3.0.1 (Singer). If $f : \mathbb{R} \rightarrow \mathbb{R}$ has $Sf < 0$ then the stable set of every attracting orbit for f either contains a critical point or is unbounded. In addition, if f has n critical points, then it admits at most $n + 2$ attracting orbits.

We briefly note that in the case of an unbounded stable set, if we restrict to a compact trapping region then this would imply that a boundary point is in the stable set. This gives the alternative conclusion that each attracting orbit has a critical point or boundary point in it's stable set.

While the negative Schwarzian derivative may seem arbitrary, it is actually directly related to a sufficient condition to guarantee the Minimum Principle for the derivative, see [Correa and Martín \(2024\)](#). In particular, $Sf < 0$ implies $|f'(x)| \geq \min\{|f'(a)|, |f'(b)|\}$ where $x \in [a, b]$. This can be used to prove Singer's theorem by controlling the derivative. In addition, a lot of work has been done under the negative Schwarzian condition and much can be concluded from this control on the derivative (e.g. distortion bounds). Many of these results can be found in [Melo and Strien \(1993\)](#).

By applying Singer's theorem we can get the following lemma.

Lemma 3.0.2. When I is a closed interval, if we replace condition (c) in 3.0.1 with the following, then $f \in \mathcal{M}$:

- (a) f unimodal with minimizer $\hat{x}$
- (b) $f(\hat{x}) < \hat{x} < f^2(\hat{x})$ and $J = [f(\hat{x}), f^2(\hat{x})]$ is a trapping region
- (c) $|(f^k)'(\hat{x})| \geq 1$ for sufficiently large k .
- (d) $f^k(I) \subset J$ for k sufficiently large

Proof.

These conditions will imply condition (c) of 3.0.1 via Singer's theorem. By Singer's theorem, it is sufficient to check that the orbit of the critical point and the endpoints is expanding since any nonrepelling periodic orbit must have one of these points in its basin of attraction. However, by condition (b) these points have the same limit set and by condition (c) we have expansion on this limit set. Thus, no nonrepelling orbit can exist so any periodic orbits must be repelling.

Condition (d) allows us to extend the domain back to $I = [a, b]$. Let $y \in I$ be a periodic point of period p and suppose it is nonrepelling. By condition (d) $y \in J$. Then by Singer's theorem, a, b or $\hat{x}$ is in the basin of attraction of the periodic orbit of y . However, since $f^k(I) \subset J$ for sufficiently large k then

$$\omega(z) = \omega(f^k(z)) = \{f^i(y) : i = 0, 1, \dots, p-1\}.$$

Since $f|_J$ has no nonrepelling periodic orbits then y must be a repelling thus a contradiction. So $f : I \rightarrow I$ has no nonrepelling periodic orbits, which is condition (c) from the previous lemma. $\square$

In regards to the above lemma, one should reference the Figure 2.1 in regards to the condition that the unimodal map has a minimum. Many of the standard texts on one dimensional unimodal maps use the convention of a maximizer (e.g. the logistic map) but our focus is on LSTM.

Next, we introduce a key definition from the theory of [Wang and Young \(2006\)](#) necessary to get the positive Lebesgue measure set of chaotic parameters.

Definition 3.0.3. A family of maps $f_a : I \rightarrow I$ for a closed interval I and $a \in [a_1, a_2]$ is called an *admissible family* if the mapping $(x, a) \mapsto f_a(x)$ is C^2 and there exists $a^* \in [a_1, a_2]$ such that

1. $f_{a^*} \in \mathcal{M}$
2. for $\hat{x} \in C(f_{a^*})$ a critical point of f_{a^*} and $q = f_{a^*}(\hat{x})$ we have

$$\frac{d}{da} (f_a(\hat{x}(a)) - q(a))|_{a=a^*} \neq 0 \quad (3.1)$$

where $\hat{x}(a)$ and $q(a)$ are continuations of $\hat{x}$ and q respectively.

We refer to 3.1 as the *parameter transversality condition*.

The parameter transversality condition intuitively means that the change in parameter changes the qualitative behavior of the map f_a near the point a^* . To see the necessity of this, consider the following extreme example where parameter transversality is not satisfied. Let $g : I \rightarrow I$ be a Misiurewicz map and consider the family φ_a of homeomorphism of I with $a \in [a_1, a_2]$. Then the family $h_a := \varphi_a^{-1} \circ g \circ \varphi_a$ is such that h_a is a Misiurewicz map for all $a \in [a_1, a_2]$. Thus, there is a positive Lebesgue measure set of chaotic parameters but in a trivial way. The parameter transversality prevents such triviality. For more information see [Wang and Young \(2006\)](#), [Wang and Young \(2008\)](#) or [Melo and Strien \(1993\)](#).

3.1 Topological Horseshoes

In this section we present two proofs that the one parameter LSTM family contains a topological horseshoe for a given set of parameters. The first method uses the technique of [Zgliczynski \(1997\)](#) which relies on an appropriate homotopy to a piecewise affine dynamical system in order to compute certain fixed point indices. The second is a direct method that mimics the proof of [Zgliczynski \(1996\)](#) and [Zgliczynski \(1997\)](#) but does not rely on any algebraic topology directly. We start by introducing the necessary prerequisites and then present each proof.

Let $\Sigma_K = \{0, 1, \dots, K-1\}^{\mathbb{N}}$ be the space of all sequences on K symbols with the Tychonoff topology. Let σ denote the shift operator $(\sigma(\gamma))_i = \gamma_{i+1}$ for $\gamma \in \Sigma_K$. Given any $K \times K$ transition matrix (or directed adjacency matrix), $A = [\alpha_{ij}]$ we can define a subspace $\Sigma_A = \{\gamma = (\gamma_i)_{i \in \mathbb{N}} : \alpha_{\gamma_i, \gamma_{i+1}} > 0\}$. These spaces are invariant under σ .

Let $f : M \rightarrow M$ be continuous. We say f has a *topological horseshoe* if there exists an invariant subset $S \subset M$ such that $f|_S$ is semiconjugate to a shift of finite type. That is, if there exists (directed) transition matrix A and mapping $\pi : S \rightarrow \Sigma_A$ such that $\pi \circ f|_S = \sigma \circ \pi$. The set S is typically defined via an isolating neighborhood N so that $S = \text{Inv}(N, f)$. In addition, A should be irreducible in the sense that for each (i, j) there exists k such that $(A^k)_{i,j} > 0$. For more information on symbolic dynamics and shifts of finite type see [appendix C.2](#).

3.1.1 Zgliczynski's Approach

Zgliczynski's approach is stated in terms of 2-dimensional maps. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$. We define a collection $\mathcal{N} = \{N_i\}_{i \in \mathcal{I}}$ of closed sets that is an isolating neighborhood which will cover our horseshoe. Zgliczynski does this by restricting to a band around the x -axis bounded by $d \leq y \leq u$. This strip is then partitioned into rectangles in the following way.

Definition 3.1.1. In [Zgliczynski \(1997\)](#), the author defines a particular isolating neighborhood to cover the horseshoe, which we will call a *Zgliczynski horseshoe cover*. Fix $u, d \in \mathbb{R}$ and a sequence $-\infty = a_{-1} < a_0 < \dots < a_{2K-2} = \infty$ where $a_i \in \mathbb{R}$ for $i = 0, 1, \dots, 2K-1$.

$$N_i = [a_{2i}, a_{2i+1}] \times [d, u]$$

$$E_i = (a_{2i-1}, a_{2i}) \times [d, u]$$

$$N = \bigcup_{i=0}^{K-1} N_i$$

$$E = \bigcup_{i=0}^K E_i.$$

Let $V(N) = \{\{a_i\} \times [d, u] : i = 0, 1, \dots, K\}$ be all the vertical edges of the N_i sets. Suppose further that for each E_i there exists E'_i such that

$$\begin{aligned} E'_i \cap (-\infty, \infty) \times [d, u] &= E_i \\ cl(E'_i) \cap (N \setminus V(N)) &= \emptyset \\ cl(E'_i) \cap cl(E'_j) &= \emptyset \quad \text{for } i \neq j. \end{aligned} \tag{3.2}$$

where E_i is a deformation retract of E'_i . That is, there exists homotopies $h_i : [0, 1] \times E'_i \rightarrow E'_i$ such that

$$\begin{aligned} h_i(0, p) &= p & \text{for } p \in E'_i \\ h_i(1, p) &\in E_i & \text{for } p \in E'_i \\ h_i(t, p) &= p & \text{for } p \in E_i. \end{aligned} \tag{3.3}$$

Let $\mathcal{N} = \{N_i\}_{i=0}^{K-1}$, $\mathcal{E} = \{E_i\}_{i=0}^K$ and $\mathcal{E}' = \{E'_i\}_{i=0}^K$. Then we call the triple $(\mathcal{N}, \mathcal{E}, \mathcal{E}')$ a *Zgliczynski horseshoe cover*.

Definition 3.1.2. Let $(\mathcal{N}, \mathcal{E}, \mathcal{E}')$ be a *Zgliczynski horseshoe cover* as above. Let $D \subset \mathbb{R}^2$ be an open set such that $N \subset D$ and $f : D \rightarrow \mathbb{R}^2$ continuous. Then f is called a *topological shift map*, (relative to $(\mathcal{N}, \mathcal{E}, \mathcal{E}')$) if there exists function $\ell, r : \{0, 1, \dots, K-1\} \rightarrow \{0, 1, \dots, K-1\}$ such that

$$\begin{aligned} f(L(N_i)) &\subset E'_{\ell(i)} \\ f(R(N_i)) &\subset E'_{r(i)} \\ f(N) &\subset E' \cup N, \end{aligned} \tag{3.4}$$

where $L(N_i)$ and $R(N_i)$ denote the left and right edge of N_i respectively. That is, $L(N_i) = \{a_{2i}\} \times [d, u]$ and $R(N_i) = \{a_{2i+1}\} \times [d, u]$.

Definition 3.1.3. A topological shift map f , has an associated transition matrix $A(f)$ defined by

$$A(f)_{i,j} = \begin{cases} 1 & \text{if } f(N_i) \supset N_j, \\ 0 & \text{else.} \end{cases}$$

Theorem 3.1.1 (Zgliczynski). Let f be a topological shift map. Then f is semi-conjugate to the shift of finite type $\sigma : \Sigma_A^+ \rightarrow \Sigma_A^+$. In addition, $\Sigma_{A(f)}^+ \subset \pi(\text{Inv}(N, f))$. The preimage of any periodic symbolic sequence from $\Sigma_{A(f)}^+$ contains periodic points of f .

It's worth mentioning that the complexity of the dynamics of $f|_{\text{Inv}(N, f)}$ depends on transition matrix A . For example, if A is the identity matrix then there are only fixed points and $\text{Inv}(N, f)$ is finite. In particular, the topological entropy of the shift $\sigma : \Sigma_A^+ \rightarrow \Sigma_A^+$ is $\log(\lambda)$ where λ is the leading eigenvalue of A (when A is irreducible).

Next we consider the one dimensional Ψ_a map trivially extended to 2 dimensions to construct a Zgliczynski horseshoe cover.

$$\begin{aligned} F(x, y) &= [0, \Psi_a(y)]^T \\ &= \begin{bmatrix} 0 \\ G(\frac{1}{2} \tanh(y), y, a) \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ \sigma(ax)y + \sigma(k_1x) \tanh(k_2x) \end{bmatrix} \end{aligned}$$

Theorem 3.1.2. Let parameters of Ψ_a be as in 4.5. Let $\mathcal{M} = \{M_i\}_{i=0}^4$ be $M_i = [-\varepsilon, \varepsilon] \times N_i$ and $M = \bigcup_{i=0}^4 M_i$. Let $\mathcal{E}$ be the connected components of the complement $([-\varepsilon, \varepsilon] \times \mathbb{R}) \setminus M$. Then $(\mathcal{M}, \mathcal{E}, \mathcal{E})$ is a Zgliczynski horseshoe cover of F . In addition, F is a topological shift map with respect to $(\mathcal{M}, \mathcal{E}, \mathcal{E})$.

Proof.

It is simple to check that $(\mathcal{M}, \mathcal{E}, \mathcal{E})$ is a Zgliczynski horseshoe cover. First note that the

homotopy condition (3.3) is trivial since $\mathcal{E} = \mathcal{E}'$. The conditions (3.2) also follows trivially from $\mathcal{E} = \mathcal{E}'$.

To show that Ψ_a is a topological shift map with respect to $(\mathcal{M}, \mathcal{E}, \mathcal{E})$ we must verify the necessary containment conditions (3.4). Since the image of F is contained in $\{0\} \times \mathbb{R}$ then

$$F(M) \subset \{0\} \times \mathbb{R} \subset E \cup M.$$

Lastly, the conditions

$$F(L(M_i)) \subset E'_{\ell(i)}$$

$$F(R(M_i)) \subset E'_{r(i)}$$

are equivalent to show that for all (i, j) either $\Psi_a(N_i) \supsetneq N_j$ or $\Psi_a(N_i) \cap N_j = \emptyset$. This is confirmed by computer-assistance in lemma 4.2.2.

The associated transition matrix and function ℓ and r are

$$A(F) = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix} \quad \begin{array}{c|cc} i & \ell & r \\ \hline 1 & 4 & 5 \\ 2 & 3 & 4 \\ 3 & 1 & 3 \\ 4 & 0 & 1 \\ 5 & 1 & 3 \end{array}$$

□

By theorem 3.1.1 π is surjective and $\Sigma_{A(F)} \subset \pi(\text{Inv}(M, F))$. Note that $\lambda \approx 1.38$ is the leading eigenvalue of $A(F)$, which means the map F has positive topological entropy.

Lastly, we show that $F|_{\text{Inv}(M, F)}$ is conjugate to $\Psi_a|_{\text{Inv}(N, \Psi_a)}$ which would prove that Ψ_a is also a topological shift map.

Lemma 3.1.1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and define $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ to be $F(x, y) = (0, f(y))$. Suppose F is a topological shift map such that $M = [-\varepsilon, \varepsilon] \times N$ is the isolating neighborhood as described above. Then $F|_{\text{Inv}(M, F)}$ is conjugate to $f|_{\text{Inv}(N, f)}$ via the inclusion map $\iota(y) = (0, y)$.

Proof.

First, we prove that $\text{Inv}(M, F) = \{0\} \times \text{Inv}(N, f)$. To see this, first note that $F(x, y) \in \{0\} \times N$ for all $(x, y) \in M$. Thus $\text{Inv}(M, F) = \text{Inv}(M, f) \cap (\{0\} \times N)$. In addition, for $i \geq 0$

$$\begin{aligned}
 F|_M^{-i}(M) \cap (\{0\} \times N) &= \{(x, y) \in M : F^i(x, y) \in M\} \cap (\{0\} \times N) \\
 &= \{(0, y) \in \{0\} \times N : F^i(0, y) \in M\} \\
 &= \{(0, y) \in \{0\} \times N : F^i(0, y) \in \{0\} \times N\} \quad \text{since } F(x, y) \in \{0\} \times N \\
 &= \{(0, y) \in \{0\} \times N : f^i(y) \in N\} \\
 &= \{0\} \times f|_N^{-i}(N).
 \end{aligned}$$

and

$$\begin{aligned}
 F|_M^i(M) &= \{(u, v) \in M : F|_M^i(x, y) = (u, v), \text{ for some } (x, y) \in M\} \\
 &= \{(0, v) \in \{0\} \times N : F|_M^i(x, y) = (0, v), \text{ for some } (x, y) \in M\} \\
 &= \{(0, v) \in \{0\} \times N : F|_M^i(0, y) = (0, v), \text{ for some } (0, y) \in \{0\} \times N\} \\
 &= \{(0, v) \in \{0\} \times N : f|_N^i(y) = v, \text{ for some } y \in N\} \\
 &= \{0\} \times f|_N^i(N).
 \end{aligned}$$

With these two equalities combined we get

$$\begin{aligned}
 \text{Inv}(M, F) &= \bigcap_{i \in \mathbb{Z}} F^{-i}(M) \\
 &= \bigcap_{i \in \mathbb{Z}} (\{0\} \times f|_N^{-i}(N)) \\
 &= \{0\} \times \bigcap_{i \in \mathbb{Z}} f|_N^{-i}(N) \\
 &= \{0\} \times \text{Inv}(N, f).
 \end{aligned}$$

Next we show that the inclusion $\iota : \text{Inv}(N, f) \rightarrow \text{Inv}(M, F)$ is a homeomorphism. Clearly, the map is a bijection so we are left to show continuity in both directions. Since both domain

and codomain are Hausdorff and compact it suffices to show continuity in one direction which holds since both spaces have the subspace topology.

Lastly, we show that ι is a conjugacy.

$$\begin{aligned} (\iota \circ f)(y) &= (0, f(y)) \\ &= F(0, y) \\ &= (F \circ \iota)(y). \end{aligned}$$

□

Applying the above lemma to the Ψ_a map we get that the following diagram commutes.

$$\begin{array}{ccccc} \text{Inv}(N, \Psi_a) & \xrightarrow{\iota} & \text{Inv}(M, F) & \xrightarrow{\pi} & \Sigma_{A(F)} \\ \Psi_a \downarrow & & \downarrow F & & \downarrow \sigma \\ \text{Inv}(N, \Psi_a) & \xrightarrow{\iota} & \text{Inv}(M, F) & \xrightarrow{\pi} & \Sigma_{A(F)} \end{array}$$

3.1.2 Direct Approach

For the direct proof, we directly construct a homeomorphism $\pi_a^{-1} : \Sigma_A \rightarrow \text{Inv}(N, f_a)$ such that $\pi_a \circ f_a \circ \pi^{-1} = \sigma$ for a family f_a under certain conditions. Then we show the the one parameter LSTM family satisfies these conditions.

Definition 3.1.4. Let $f : I \rightarrow I$ be a continuous map on some interval I . Let $\mathcal{N} = \{N_i\}_{i=0}^{K-1}$ be a collection of closed intervals in I such that for all i, j ,

$$f(N_i) \supsetneq N_j \quad \text{or} \quad f(N_j) \cap N_i = \emptyset$$

and $\partial(f(N_i)) \cap N = \emptyset$. Such a collection $\mathcal{N}$ is called a *horseshoe cover*. We define the associated transition matrix $A(f)$ by

$$A(f)_{ij} = \begin{cases} 1 & \text{if } f(N_i) \supsetneq N_j \\ 0 & \text{else.} \end{cases}$$

Theorem 3.1.3. Let $\mathcal{N} = \{N_i\}_{i=0}^{K-1}$ be a *horseshoe cover*. In addition, assume that $|f'| > 1$ on $N = \bigcup_{i=1}^{K-1} N_i$ and let A be the associated transition matrix. Then $f|_{\text{Inv}(N, f)}$ is conjugate to a shift of finite type. That is, there exists a homeomorphism $\pi : \text{Inv}(N, f) \rightarrow \Sigma_A$ such that $\pi^{-1} \circ \sigma \circ \pi = f$.

Proof.

First we define the mapping π^{-1} and show that it is well-defined. We will later show that it is a homeomorphism. For a given sequence $\gamma \in \Sigma_A$ we define $\pi^{-1} : \Sigma_A \rightarrow \text{Inv}(N, f)$ by

$$\pi^{-1}(\gamma) = \bigcap_{i=0}^{\infty} f^{-i}(N_{\gamma_i}).$$

To show that it is well defined, is to show that the right hand side is in fact a single point. To do this, we will first write an equivalent expression for the right hand side.

For each (i, j) such that $A(i, j) > 0$ we have $f|_{N_i}(N_i) \supset N_j$. Since $f|_{N_i}$ is strictly monotone (injective) then we can define a right inverse $g_{j,i} : N_j \rightarrow N_i$ such that $g_{j,i}(x) := (f|_{N_i})^{-1}(x)$. That is $f|_{N_i} \circ g_{j,i} = id_{N_j}$. Then define

$$M_k := \bigcap_{i=0}^k f^{-i}(N_{\gamma_i}) = g_{\gamma_1, \gamma_0} \circ \dots \circ g_{\gamma_{k+1}, \gamma_k}(N_{\gamma_{k+1}}). \quad (3.5)$$

Let α be such that $|(f|_N)'| \geq \alpha > 1$. Then $|g'_{j,i}| < 1/\alpha < 1$. Thus, we have

$$m\left(\bigcap_{i=0}^k f^{-i}(N_{\gamma_i})\right) \leq m(N_{\gamma_k})/\alpha^k \leq \max\{m(N_i)\}_{i=1}^{K-1}/\alpha^k$$

Thus, in the limit, the measure must go to zero. In addition, by (3.5), we know that the M_k is connected for all k . Thus the limit $\lim_{k \rightarrow \infty} M_k$ is connected with measure zero so it must be a single point. Thus, π^{-1} is well-defined. In fact, π^{-1} is a bijection since each point only has one symbolic dynamics and that sequence must be in Σ_A .

Next, we show that this map is a homeomorphism. Since Σ_A and $\text{Inv}(N, f)$ are both compact and Hausdorff then it suffices to show continuity in one direction. Let $U \subset \text{Inv}(N, f)$ be an open set and let $\gamma \in \pi(U)$. Then $\pi^{-1}(\gamma) \in U$. Let $\delta > 0$ such that $B_\delta(\pi^{-1}(\gamma)) \cap \text{Inv}(N, f) \subset U$. Thus, if we can find an open set $V \subset \Sigma_A$ such that $\gamma \in V$ and $\pi^{-1}(V) \subset$

$B_\delta(\pi^{-1}(\gamma))$ then $\gamma \in V \subset \pi(U)$ and our proof is complete. Let $M \in \mathbb{N}$ be such that

$$\max\{m(N_i)\}_{i=1}^{K-1}/\alpha^M < \delta/3$$

and define $V = \{\varphi \in \Sigma_A : \varphi_i = \gamma_i, i = 0, 1, \dots, M\}$. Then V is open in the Tychonoff topology, $\gamma \in V$ and

$$\pi^{-1}(V) \subset \bigcap_{i=1}^M f^{-i}(N_{\gamma_i}) \subset B_\delta(\pi^{-1}(\gamma)).$$

This completes the proof. □

Theorem 3.1.4. Consider the one dimensional family Ψ_a with parameters as in 4.5. Then for each a , $\Psi_a|_{\text{Inv}(N, \Psi_a)}$ is conjugate to the subshift of finite type $\sigma : \Sigma_A^+ \rightarrow \Sigma_A^+$ where A is as in the proof of theorem .

Proof.

This theorem is the result of two computer-assisted lemmas. Lemma 4.2.2 verifies the necessary containment conditions for $\mathcal{N}$ to be a horseshoe cover with respect to $\Psi_a : I \rightarrow I$ for all a . Then lemma 4.2.1 verifies that $|\Psi_a|_N| > 1$. Thus our result is true by theorem . □

3.2 Continuations and Hitting the Horseshoe

In this section we prove the existence of a unique continuation of horseshoe points and the critical point for certain parameters. This continuity is then used to argue that there exists countably many parameters a^* such that $\Psi_{a^*}^n(\hat{y}(a^*)) \in \text{Inv}(N, \Psi_{a^*})$ for n sufficiently large, where $\hat{y}(a)$ denotes the continuation of the unique critical point of Ψ_a . Thus for certain parameters, the critical orbit gets trapped in the horseshoe.

3.2.1 Continuation of Horseshoe Points

Theorem 3.2.1. Let $f_a : I \rightarrow I$ be a continuous family of dynamical systems with $a \in [a_1, a_2]$ such that $\mathcal{N} = \{N_i\}_{i=0}^{K-1}$ is a horseshoe cover for all a . Suppose further that $|f'_a| > 1$

for all a . Then $A(f_a)$ is constant in a and for each $\gamma = \{\gamma_n\} \in \Sigma_{A(f_a)} = \Sigma_{A(f)}$ the continuation $\pi^{-1}(\gamma) : [a_1, a_2] \rightarrow I$ is continuous.

Proof.

First, the fact that $A(f_a)$ is constant in a follows quickly from the fact that $\mathcal{N}$ is a horseshoe cover for each f_a since the transition matrix is determined by the covering conditions, which are constant in a . Let $A = A(f_a)$. In addition, let $\pi^{-1} : [a_1, a_2] \times \Sigma_A^+ \rightarrow I$ be such that $\pi^{-1}(a, \cdot) : \Sigma_A^+ \rightarrow \text{Inv}(N, f_a)$ is the conjugacy from theorem 3.1.2.

Next, we prove the continuity of the continuation. Let $g_{a,j,i} : N_j \rightarrow N_i$ be the right inverse of $f_a|_{N_i} : N_i \rightarrow I$ as in the proof of theorem 3.1.2. Let α be such that $|(f_a|_N)'| > \alpha > 1$ for all a .

To show continuity of $\pi^{-1}(\gamma) : [a_1, a_2] \rightarrow I$ for all $\gamma \in \Sigma_A^+$, suppose there exists $\gamma \in \Sigma_A$, $a \in [a_1, a_2]$ and $\varepsilon > 0$ such that

$$|\pi^{-1}(\gamma)(a) - \pi^{-1}(\gamma)(a + \delta)| > \varepsilon$$

for all δ sufficiently small. Let

$$M_{a,k} := \bigcap_{i=0}^k f_a^{-i}(N_{\gamma_i}) = g_{a,\gamma_1,\gamma_0} \circ \dots \circ g_{a,\gamma_{k+1},\gamma_k}(N_{\gamma_{k+1}}).$$

so that $\pi_a^{-1}(\gamma) = \lim_{k \rightarrow \infty} M_{a,k}$.

Then for all a , we have

$$m(M_{a,k}) \leq m(N_{\gamma_k})/\alpha^k \leq \max\{m(N_i)\}_{i=1}^{K-1}/\alpha^k.$$

Since $\alpha > 1$, we can pick k to be sufficiently large such that $m(M_{a,k}) < \varepsilon/3$ for all a .

Then for all δ ,

$$\begin{aligned} \pi^{-1}(\gamma)(a) &\in M_{a,k} = g_{a,\gamma_0,\gamma_1} \circ \dots \circ g_{a,\gamma_{n-1},\gamma_n}(N_{\gamma_n}) \\ \pi^{-1}(\gamma)(a + \delta) &\in M_{a+\delta,k} = g_{a+\delta,\gamma_0,\gamma_1} \circ \dots \circ g_{a+\delta,\gamma_{n-1},\gamma_n}(N_{\gamma_n}) \end{aligned}$$

By continuity in a , the limit

$$\lim_{\delta \rightarrow 0} g_{a+\delta, \gamma_0, \gamma_1} \circ \dots \circ g_{a+\delta, \gamma_n, \gamma_{n-1}}(N_{\gamma_n}) = g_{a, \gamma_1, \gamma_0} \circ \dots \circ g_{a, \gamma_n, \gamma_{n-1}}(N_{\gamma_n})$$

But this contradicts the fact that $|\pi^{-1}(\gamma)(a) - \pi^{-1}(\gamma)(a + \delta)| > \varepsilon$. Thus, $\pi^{-1}(\gamma) : [a_1, a_2] \rightarrow I$ is continuous. $\square$

Theorem 3.2.2. Consider the one-dimensional family Ψ_a with parameters as in 4.5. Then $\mathcal{N}$ is a horseshoe cover for all a . In addition, for $\gamma \in \Sigma_A^+$ the mapping $\pi^{-1}(\gamma) : [a_1, a_2] \rightarrow N$ is continuous.

Proof.

The family Ψ_a is continuous in a by inspection of the formula. Lemma 4.2.2 shows that $\mathcal{N}$ is a horseshoe cover for all a . Lemma 4.2.1 shows that $|\Psi'_a| > 1$ on N for all a . Thus by theorem 3.2.1 we have the continuity of $\pi^{-1} : [a_1, a_2] \rightarrow N$. Thus horseshoe points have a well-defined continuation. $\square$

Lemma 3.2.1. Consider the one-dimensional family Ψ_a with parameters as in 4.8. Then for each a there is a unique critical point $\hat{y}(a)$ of Ψ_a . Furthermore, the mapping $\hat{y} : [a_1, a_2] \rightarrow I$ is continuous.

Proof.

Let $g : \mathbb{R} \times [a_1, a_2] \rightarrow \mathbb{R}$ be defined by $g(y, a) = \frac{d}{dy} \Psi_a(y)$. Then 0 is a regular value of g since

$$\begin{aligned} Dg(y_0, a_0) &= \begin{bmatrix} g_y(y_0, a_0) & g_a(y_0, a_0) \end{bmatrix} \\ &= \begin{bmatrix} \frac{\partial^2}{\partial y^2} \Psi_a(y)|_{y_0, a_0} & \frac{\partial^2}{\partial y \partial a} \Psi_a(y)|_{y_0, a_0} \end{bmatrix} \end{aligned}$$

and for each (y_0, a_0) such that $g(y_0, a_0), \frac{\partial^2}{\partial y^2} \Psi_a(y)|_{y_0, a_0} > 0$ by lemma 4.2.3. Thus, $g^{-1}(0)$ is a smooth submanifold (possibly with boundary) of $\mathbb{R} \times [a_1, a_2]$. For each $(y_0, a_0) \in g^{-1}(0)$ we may define a local function $\hat{y}(a)$ such that $g(\hat{y}(a), a) = 0$ for a is some small neighborhood

of a_0 . And

$$\begin{aligned}\frac{d}{da}g(y, a) &= g_y(y, a)\frac{dy}{da} + g_a(y, a) \\ 0 &= g_y(\hat{y}(a), a)\frac{d\hat{y}}{da} + g_a(\hat{y}(a), a) \\ \frac{d\hat{y}}{da} &= \frac{-g_a(\hat{y}(a), a)}{g_y(\hat{y}(a), a)}.\end{aligned}$$

Since $g_y(\hat{y}(a), a) > 0$ by lemma 4.2.3, then $|\frac{d\hat{y}}{da}| \neq \infty$. This implies that no connected component of $g^{-1}(0)$ is homeomorphic to a circle. In addition, by lemma 4.2.3, there exists $a' \in [a_1, a_2]$ such that $g(y, a') = 0$ has a unique solution. Thus $g^{-1}(0)$ has only one connected component. Therefore, $g^{-1}(0)$ is the graph of a function which we denote $\hat{y}(a)$. $\square$

3.2.2 Hitting the Horseshoe

Our next goal is to show that for certain parameters, Ψ_a has no nonrepelling periodic orbits. To do this, we show the existence of countably many parameters such that the critical orbit lands in the horseshoe.

Theorem 3.2.3. Consider the one-dimensional family Ψ_a with parameters as in 4.10. Then there exists countably many parameters $a^* \in [a_1, a_2]$ such that $\Psi_{a^*}^n(\hat{y}(a^*)) \in \text{Inv}(N, \Psi_{a^*})$ for n sufficiently large.

Proof.

By lemma 4.4.1 we have the existence of i and n such that

$$N_i \subset \text{convexhull}(\{\Psi_{a_j}^n(\hat{y}(a_j)) : j = 1, 2\})$$

Without loss of generality, assume that $\Psi_{a_1}^n(\hat{y}(a_1)) \leq \Psi_{a_2}^n(\hat{y}(a_2))$. Then we have for any a and $\gamma \in \Sigma_A^+$ such that $\gamma_0 = i$,

$$\Psi_{a_1}^n(\hat{y}(a_1)) \leq \underline{N_i} \leq \pi^{-1}(\gamma)(a) \leq \overline{N_i} \leq \Psi_{a_2}^n(\hat{y}(a_2)).$$

By lemma 3.2.1 and theorem 3.2.2 we have that $\Psi_a^n(\hat{y}(a))$ and $\pi^{-1}(\gamma)(a)$ are both continuous. Thus by the intermediate value theorem there exists $a^* \in [a_1, a_2]$ such that $\Psi_{a^*}^n(\hat{y}(a^*)) =$

$\pi^{-1}(\gamma)(a^*) \in \text{Inv}(N, \Psi_{a^*})$. Since there are countable many sequences γ such that $\gamma_0 = i$ then there are countably many such a^* . $\square$

Our next goal is to prove that the parameters a^* from the above theorem are Misiurewicz points, that is $\Psi_{a^*} \in \mathcal{M}$.

Theorem 3.2.4. Consider the one-dimensional family Ψ_a with parameters as in 4.10 and let $a^* \in [a_1, a_2]$ be as in theorem 3.2.3. Then $\Psi_{a^*} \in \mathcal{M}$.

Proof.

We will show that Ψ_{a^*} satisfies the conditions of lemma 3.0.1 and 3.0.2. Conditions (a) and (b) of lemma 3.0.1 are satisfied by computer assistance in proposition 4.3.1 and 4.2.3 respectively. Condition (d) of lemma 3.0.1 is true since $\Psi_{a^*}^n(\hat{y}(a^*)) \in \text{Inv}(N, \Psi_{a^*})$ for n sufficiently large by theorem 3.2.3. We are left to confirm (c) of lemma 3.0.1 which can be replaced by lemma 3.0.2. Condition (c) of lemma 3.0.2 is true by the fact that $\Psi_{a^*}^n(\hat{y}(a^*)) \in \text{Inv}(N, \Psi_{a^*})$ for n sufficiently large. Condition (a) of lemma 3.0.2 is true by lemma 3.2.1 and 4.2.3. Conditions (b) and (d) of lemma 3.0.2 are verified by computer assistance in lemmas 4.4.2. Thus $\Psi_{a^*} \in \mathcal{M}$. $\square$

3.3 One Dimensional Admissible Family

In this section, we verify that our one parameter LSTM family satisfies the parameter transversality of D.0.2. Then theorem 1.0.2 follows from D.0.2.

3.3.1 Parameter Transversality

In this section, we prove that Ψ_a is an admissible family as in Wang and Young (2006). Then by D.0.2, we prove the existence of absolutely continuous invariant measures for the map Ψ_a for a positive Lebesgue measure set of parameters. The only condition to verify in order to apply theorem D.0.2 is the parameter transversality condition.

Theorem 3.3.1. Consider the one parameters LSTM family Ψ_a with parameters as in 4.11 and let a^* be such that $\Psi_{a^*} \in \mathcal{M}$ as in theorem 3.2.4. Let $q = \Psi_{a^*}(\hat{y}(a^*))$ and $q(a)$ be its

continuation. Then

$$\frac{d}{da} (\Psi_a(\hat{y}(a)) - q(a))|_{a=a^*} \neq 0.$$

Proof.

Equivalently, we show that

$$\sum_{i=0}^{\infty} \frac{\partial_a f_a(\hat{x}_i(a^*))|_{a=a^*}}{(f_{a^*}^i)'(\hat{x}_1(a^*))} \neq 0. \quad (3.6)$$

The above sum is nonzero if there exists some N such that

$$\left| \sum_{i=0}^K \frac{\partial_a f_a(\hat{x}_i(a^*))|_{a=a^*}}{(f_{a^*}^i)'(\hat{x}_1(a^*))} \right| > \sum_{i=K+1}^{\infty} \left| \frac{\partial_a f_a(\hat{x}_i(a^*))|_{a=a^*}}{(f_{a^*}^i)'(\hat{x}_1(a^*))} \right|.$$

For LSTM we have

$$\begin{aligned} \partial_a \Psi_a(y) &= \sigma' \left(a \frac{1}{2} \tanh(y) \right) y \frac{1}{2} \tanh(y) \\ |\partial_a \Psi_a(y)| &= \sigma' \left(a \frac{1}{2} \tanh(y) \right) y \frac{1}{2} \tanh(y) \\ &\leq \left| \sigma' \left(a \frac{1}{2} \tanh(y) \right) \right| |y| \frac{1}{2} |\tanh(y)| \\ &\leq \frac{1}{4} \cdot \frac{1}{2} \cdot \frac{1}{2} \\ &= 1/16. \end{aligned}$$

By theorem 3.2.3 we know that there exists K such that $\Psi_{a^*}^k(\hat{y}(a^*)) \in \text{Inv}(N, \Psi_{a^*})$ for $k \geq K$. Suppose that K is such an integer. Let α be such that $|(\Psi_{a^*}|_N)'| > \alpha > 1$. Then for any point $z \in \text{Inv}(N, \Psi_{a^*})$ we have $|\Psi_{a^*}'(z)| > \alpha$. Thus,

$$\begin{aligned} \left| \sum_{i=K+1}^{\infty} \frac{\partial_a \Psi_a(\hat{y}_i(a^*))|_{a=a^*}}{(\Psi_{a^*}^i)'(\hat{y}_1(a^*))} \right| &\leq \sum_{i=K+1}^{\infty} \left| \frac{\partial_a \Psi_a(\hat{y}_i(a^*))|_{a=a^*}}{(\Psi_{a^*}^i)'(\hat{y}_1(a^*))} \right| \\ &\leq \frac{1}{16} \sum_{i=K+1}^{\infty} \frac{1}{|(\Psi_{a^*}^i)'(\hat{y}_1(a^*))|} \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{16 |(\Psi_{a^*}^K)'(\hat{y}(a^*))|} \sum_{i=K+1}^{\infty} \left| \frac{1}{\left(\prod_{j=K}^{i-1} \Psi'_{a^*}(\Psi^j(\hat{y}_1(a^*))) \right)} \right| \\
&\leq \frac{1}{16 |(\Psi_{a^*}^K)'(\hat{y}(a^*))|} \sum_{i=K+1}^{\infty} \frac{1}{|\alpha^{i-K}|} \\
&\leq \frac{1}{16 |(\Psi_{a^*}^K)'(\hat{y}(a^*))|} \frac{1}{1 - \alpha^{-1}}
\end{aligned}$$

Then the following inequality would also imply that (3.6) is true for the LSTM model.

$$\left| \sum_{i=0}^K \frac{\partial_a \Psi_a(\hat{y}_i(a^*))|_{a=a^*}}{(f_{a^*}^i)'(\hat{y}_1(a^*))} \right| > \frac{1}{16 |(f_{a^*}^K)'(\hat{y}_1(a^*))|} \frac{1}{1 - \alpha^{-1}}.$$

As per 4.5.1, this inequality is true on a small neighborhood containing the critical curve $\hat{y}(a)$ for $a \in [30.993, 31.078]$. Thus, the inequality holds when $a = a^*$. $\square$

Chapter 4

Numerical Methods and Analysis

In this section, we describe the numerical methods used to confirm the necessary lemmas used in the previous section. We include here for reference a collection of standard parameter sets that we use. We will note when a parameter can be extended to a larger set.

$$[a_1, a_2] = [30.993, 31.078] \tag{4.1}$$

$$[b_1, b_2] = [-0.0001, 0.0001]$$

$$k_1 = -6.105$$

$$k_2 = -12.21$$

$$D \times I = [-0.005, 0.005] \times [-0.165, 0.5]$$

$$\begin{aligned} \mathcal{N} &= \{N_1, N_2, \dots, N_5\} \\ &= \{[-0.11, -0.087], [-0.03, -0.014], [-0.01, 0.025], [0.038, 0.06], [0.258, 0.31]\} \end{aligned}$$

The main strategy we will use to approximate the LSTM model and its derivatives will be to consider refinements of a natural interval extension. These formulae can be found in appendix [B](#). When we use a refinement in the following sections, we'll state a length $\Delta\alpha$ for $\alpha \in \{x, y, a, b\}$ and the corresponding range. Then, the refinement takes place over a partition of the product space for each of the variables/parameters. For a function $f_{a,b} : D \times I \rightarrow D \times I$, and a interval extension F , we write

$$F_{\mathcal{P}}(D, I; [a_1, a_2], [b_1, b_2]) := \bigcap_{[x] \times [y] \times [a] \times [b] \in \mathcal{P}} F([x], [y]; [a], [b])$$

for the refinement with respect to the partition $\mathcal{P}$. Then the following inclusion tells us our interval extension is indeed an upper bound of the quantities we want to calculate:

$$f(D, I; [a_1, a_2], [b_1, b_2]) \subset F_{\mathcal{P}}(D, I; [a_1, a_2], [b_1, b_2]) = \bigcap_{[x] \times [y] \times [a] \times [b] \in \mathcal{P}} F([x], [y]; [a], [b])$$

Therefore, when we want to verify a containment $f(D, I; [a_1, a_2], [b_1, b_2]) \subset V$, for some $V \subset D \times I$, it suffices to show that $F([x], [y]; [a], [b]) \subset V$ for each partition element $[x] \times [y] \times [a] \times [b]$. This will be the main strategy in the following sections. Another slight alternative to this is to show $f(D, I; [a_1, a_2], [b_1, b_2]) \cap V = \emptyset$. This can similarly be done by checking disjointness over the partition.

Verification of disjointness can be used to verify inequalities. For another function g , if we wish to verify the inequality

$$f(D, I; [a_1, a_2], [b_1, b_2]) < g(D, I; [a_1, a_2], [b_1, b_2])$$

then we can show

$$(-\infty, \overline{f(D, I; [a_1, a_2], [b_1, b_2])}] \cap [\underline{g(D, I; [a_1, a_2], [b_1, b_2])}, \infty) = \emptyset$$

4.1 Well Definedness of the family $\Psi_{a,b}$

In this section, we show the well-definedness of our two LSTM families. We consider the parameters

$$D = [-0.005, 0.005] \tag{4.2}$$

$$I = [-0.165, 0.5]$$

$$[a_1, a_2] = [29.0, 31.2]$$

$$[b_1, b_2] = [-0.0001, 0.0001]$$

$$k_1 = -6.105$$

$$k_2 = -12.21$$

Lemma 4.1.1. Define

$$\Psi_{a,b}(x, y) = \begin{bmatrix} (\sigma(b(x + \frac{1}{2} \tanh(y)))) - \frac{1}{2} \tanh(G(x + \frac{1}{2} \tanh(y), y, a)) \\ G(x + \frac{1}{2} \tanh(y), y, a) \end{bmatrix} \quad (4.3)$$

$$G(x, y, a) = \sigma(ax)y + \sigma(k_1x) \tanh(k_2x).$$

as in equation (2.3). For parameters as in 4.2, we have $\Psi(D, I; [a_1, a_2], [b_1, b_2]) \subset D \times I$.

Proof.

Define a regular partition of the space $D \times I \times [a_1, a_2] \times [b_1, b_2]$ denoted $\mathcal{P}$ generated by $\Delta x = 0.0001, \Delta y = 0.001, \Delta a = 0.2, \Delta b = 0.0001$.

We can verify the desired containment by checking the containment $\Psi([x], [y]; [a], [b]) \subset D \times I$ for each partition element $[x] \times [y] \times [a] \times [b] \in \mathcal{P}$. The script `LemmaWellDefined.py` checks this containment for each partition element. $\square$

Lemma 4.1.2. Define $\Psi : I \times [a_1, a_2] \rightarrow I$ by

$$\Psi_a(y) = \Psi(y; a) = G\left(\frac{1}{2} \tanh(y), y, a\right) \quad (4.4)$$

$$G(x, y, a) = \sigma(ax)y + \sigma(k_1x) \tanh(k_2x).$$

as in equation (4.4). For parameters as in 4.2, we have

Then

$$\Psi(I; [a_1, a_2]) \subset I.$$

Proof.

This follows from lemma 4.1.1 since

$$\Psi(0, y, a, 0) = \begin{bmatrix} 0 \\ G\left(\frac{1}{2} \tanh(y), y, a\right) \end{bmatrix} \begin{bmatrix} 0 \\ \Psi(y, a). \end{bmatrix}$$

□

4.2 Topological Horseshoe

In this section, we present the computer-assisted lemmas used to confirm the necessary conditions, via theorems of Zgliczynski (1997) and our direct method, to prove the existence of a topological horseshoe in the LSTM model. These results in this section use the following parameter values:

$$[a_1, a_2] = [30.993, 31.078] \tag{4.5}$$

$$k_1 = -6.105$$

$$k_2 = -12.21$$

$$N_0 = [-0.11, -0.087]$$

$$N_1 = [-0.03, -0.014]$$

$$N_2 = [-0.01, 0.025]$$

$$N_3 = [0.038, 0.06]$$

$$N_4 = [0.258, 0.31]$$

Lemma 4.2.1. For the collection $\mathcal{N} = \{N_i\}_{i=0}^4$ above, and parameters as in 4.5, we have that $\Psi_a : I \rightarrow I$ is piecewise monotonic when restricted to $N = \bigcup_{i=0}^4 N_i$. In addition, $|\Psi'_a|_N| > 1$.

Proof.

It is enough to confirm the containment $\Psi'(N_i, [a_1, a_2]) \subset [1, \infty)$ or $\Psi'(N_i, [a_1, a_2]) \subset (-\infty, -1]$ for all $i = 0, 1, \dots, 4$. This is checked directly in script `LemmaHorseshoeCover.py`.

□

Lemma 4.2.2. For Ψ with parameters as in 4.5 and collection $\mathcal{N}$ we have

$$\Psi(N_i, [a_1, a_2]) \supset N_j \quad \text{xor} \quad \Psi(N_i, [a_1, a_2]) \cap N_j = \emptyset.$$

for all pairs (i, j) .

Proof.

By 4.2.1 we know that Ψ is monotone in y when restricted to some N_i for $i = 0, 1, \dots, 4$.

Then we can compute upper and lower bounds of the image by lemma A.2.1

$$\begin{aligned} \Psi(N_i, [a_1, a_2]) \supset I(N_i, [a_1, a_2]) &:= \begin{cases} [\Psi(\overline{N_i}, [a_1, a_2]), \overline{\Psi(N_i, [a_1, a_2])}] & \text{if } \Psi'_a|_{N_i} > 0 \\ [\Psi(\underline{N_i}, [a_1, a_2]), \overline{\Psi(\overline{N_i}, [a_1, a_2])}] & \text{if } \Psi'_a|_{N_i} < 0 \end{cases} \\ \Psi(N_i, [a_1, a_2]) \subset J(N_i, [a_1, a_2]) &:= \begin{cases} [\Psi(\underline{N_i}, [a_1, a_2]), \overline{\Psi(\overline{N_i}, [a_1, a_2])}] & \text{if } \Psi'_a|_{N_i} > 0 \\ [\Psi(\overline{N_i}, [a_1, a_2]), \overline{\Psi(\underline{N_i}, [a_1, a_2])}] & \text{if } \Psi'_a|_{N_i} < 0 \end{cases} \end{aligned}$$

Then the script `LemmaHorseshoeCover.py` checks the containment

$$I(N_i, [a_1, a_2]) \supset N_j \quad \text{xor} \quad J(N_i, [a_1, a_2]) \cap N_j = \emptyset.$$

This suffices to prove our claim since

$$\begin{aligned} \Psi(N_i, [a_1, a_2]) \supset I(N_i, [a_1, a_2]) \supset N_j \\ \Psi(N_i, [a_1, a_2]) \cap N_j \subset J(N_i, [a_1, a_2]) \cap N_j = \emptyset \end{aligned}$$

□

Next, we confirm some lemmas regarding the continuation of the critical point $\hat{y}(a)$.

$$[a_1, a_2] = 30.8 \tag{4.6}$$

$$k_1 = -6.105$$

$$k_2 = -12.21$$

(4.7)

Lemma 4.2.3. Let parameters be as in 4.6. There exists a neighborhood $U \subset I \times [a_1, a_2]$ such that if $\Psi'_a(y) = 0$ then $(y, a) \in U$ and $\Psi''|_U > \alpha > 0$. In addition, there exists $a' \in [a_1, a_2]$ such that $\Psi'_{a'}(y) = 0$ as a unique solution.

Proof.

Generate a partition of $I \times [a_1, a_2]$ denoted $\mathcal{P}$ generated by $\Delta y = 0.0001$ and $\Delta a = 0.05$. Let $C([y]; [a])$ be an upperbound of $\Psi'([y]; [a])$ determined by the natural interval extension of the derivative. Then we verify that if $0 \in C([y]; [a])$ then $\Psi''([y]; [a]) \subset [1e-06, \infty)$. The script `LemmaSecondDerivative.py` verifies this implication. The set U is constructed as

$$U = \bigcup \{[y] \times [a] \in \mathcal{P} : 0 \in C([y]; [a])\}$$

The script `LemmaSecondDerivative.py` confirms that for $a' = (a_1 + a_2)/2$, $\Psi_{a'}$ has a unique critical point. $\square$

4.3 Schwarzian Derivative of LSTM

In this section, we discuss the computer-assisted proof that $S(LSTM_a) < 0$. We consider parameters to be:

$$I = [-0.165, 0.5] \tag{4.8}$$

$$[a_1, a_2] = 30.8$$

$$k_1 = -6.105$$

$$k_2 = -12.21$$

Lemma 4.3.1. For parameters as in 4.8, $S(\Psi_a) < 0$.

Proof.

First note that if

$$f'''(x)f'(x) - \frac{3}{2}f''(x)^2 < 0. \tag{4.9}$$

then whenever $f'(x) \neq 0$, $Sf < 0$ (by dividing by $(f'(x))^2$).

We prove this result by showing that for a partition $\mathcal{P}$ of $I \times [a_1, a_2]$ generated by $\Delta y = 0.0001$ and $\Delta a = 0.05$, that inequality (4.9) holds on each partition element. In particular, we test

$$\Psi'''([y]; [a])\Psi'([y]; [a]) - \frac{3}{2}(\Psi([y]; [a])^2 \subset (-\infty, 0).$$

The script `LemmaNegativeSchwarzian.py` verifies this containment.

□

4.4 Critical Orbit Lands in Horseshoe

In this section, we present the necessary lemmas in the proof of 3.2.3. The results in this section use the following parameters:

$$I = -0.165 \tag{4.10}$$

$$[a_1, a_2] = [30.993, 31.078]$$

$$k_1 = -6.105$$

$$k_2 = -12.21$$

$$N_0 = [-0.11, -0.087]$$

$$N_1 = [-0.01, 0.025]$$

$$N_2 = [0.258, 0.31]$$

$$N_3 = [0.038, 0.06]$$

$$N_4 = [-0.03, -0.014]$$

Lemma 4.4.1. Let parameters be as in 4.10. Then there exists an $n \in \mathbb{N}$ and some $i \in \{0, \dots, 4\}$ such that

$$N_i \subset \text{convexhull}(\Psi_{a_1}^n(\hat{y}(a_1)), \Psi_{a_2}^n(\hat{y}(a_2))).$$

Proof.

It is sufficient to show that

$$\begin{aligned}\overline{\Psi(\hat{Y}(a_1), a_1)} \cap [N_1, \infty) &= \emptyset \\ \overline{\Psi(\hat{Y}(a_2), a_2)} \cap (-\infty, \overline{N_1}] &= \emptyset\end{aligned}$$

where $\hat{Y}(a_i)$ is an interval containing the critical point $\hat{y}(a_i)$ for $i = 1, 2$. We construct the interval $\hat{Y}(a_i)$ using interval Newton's method. The above disjointness is then confirmed by `LemmaHitHorseshoe.py`. $\square$

Lemma 4.4.2. Let parameters be as in 4.10. Then $\Psi_a(\hat{y}(a)) < \hat{y}(a) < \Psi_a^2(\hat{y}(a))$ and there exists k such that

$$\Psi_a^k(I \setminus [\Psi_a(\hat{y}(a)), \Psi_a^2(\hat{y}(a))]) \subset [\Psi_a(\hat{y}(a)), \Psi_a^2(\hat{y}(a))].$$

Proof.

Let $\mathcal{P}$ be a partition of $[a_1, a_2]$ generated by $\Delta a = 0.01$. Then for each $[a] \in \mathcal{P}$ we can approximate the critical point with an interval $\hat{Y}([a])$ using interval Newton's method, and thus approximate the images with $\Psi(\hat{Y}([a]), [a])$ and $\Psi^2(\hat{Y}([a]), [a])$. It is then simple to verify the desired containment,

$$\Psi^k(I \setminus [\overline{\Psi(\hat{Y}([a]), [a])}, \overline{\Psi^2(\hat{Y}([a]), [a])}]) \subset [\overline{\Psi(\hat{Y}([a]), [a])}, \overline{\Psi^2(\hat{Y}([a]), [a])}].$$

which is done in script `LemmaCriticalTrappingRegion.py`. $\square$

4.5 Parameter Transversality

$$\begin{aligned}I &= [-0.165, 0.5] & [a_1, a_2] &= [30.993, 31.078] \\ k_1 &= -6.105 & k_2 &= -12.21\end{aligned}\tag{4.11}$$

Lemma 4.5.1. Let Ψ_a be the one parameter family with parameters in 4.11. Then, there exists a neighborhood $U \subset I \times [a_1, a_2]$ such that $\{(\hat{y}(a), a) : a \in [a_1, a_2]\} \subset U$ and an integer N such that for $(y_0, a_0) \in U$

$$\left| \sum_{i=0}^N \frac{\partial_a \Psi_a(y)|_{a=a_0}}{(\Psi_{a_0}^i)'(y_1)} \right| > \frac{1}{16 |(\Psi_{a_0}^N)'(y_1)|} \frac{1}{1 - \alpha^{-1}} \quad (4.12)$$

where $y_1 = \Psi_{a_0}(y_0)$ and $\alpha(a) \leq \min\{\Psi'_a(y) : y \in N\}$.

Proof.

First we construct the set U . Let $\mathcal{P}$ be a regular partition of $[a_1, a_2]$ generated by $\Delta a = 1e - 05$. Then we construct U as

$$U = \bigcup_{[a] \in \mathcal{P}} [a] \times \hat{Y}([a])$$

where $\hat{y}([a]) \subset \hat{Y}([a])$. Here, $\hat{Y}([a])$ is constructed using interval Newton's method to approximate the location of the critical points of $\Psi(\cdot, [a])$.

To verify the inequality (4.13) we compute each factor in the following manner

$$\begin{aligned} \frac{\partial}{\partial a} \Psi_a(\hat{Y}([a])) &= \frac{1}{2} \tanh(\hat{Y}([a])\sigma') \left([a] \frac{1}{2} \tanh(\hat{Y}([a])) \right) \hat{Y}([a]) \\ \Psi^{(k)}(\hat{Y}([a]), [a]) &= \Psi^{(k-1)}(\hat{Y}([a]), [a]) \cdot \Psi'(\Psi^k(\hat{Y}([a]), [a]), [a]) \end{aligned}$$

These formulas come from B.13 and the chain rule respectively. Then `LemmaParameterTransversality.py` verifies the inequality

$$\left| \sum_{i=0}^N \frac{\partial_a \Psi_a(\hat{Y}([a]))|_{a=a_0}}{(\Psi_{a_0}^i)'(\Psi(\hat{Y}([a]), [a]))} \right| > \frac{1}{16 |(\Psi_{a_0}^N)'(\Psi(\hat{Y}([a]), [a]))|} \frac{1}{1 - \alpha^{-1}} \quad (4.13)$$

for each partition element $[a] \in \mathcal{P}$ and $N = 12$.



Chapter 5

A Conjecture for the Two Parameter Family $\Psi_{a,b}$

In this section, we present the conjecture 1.0.1 for the two parameter family $\Psi_{a,b}$ as in (2.3). When we let $b = 0$, then $\Psi_{a,0}(0, y) = (0, \Psi_a(y))$ and thus we can think of $\Psi_a(y)$ as the singular limit of the family $\Psi_{a,b}(x, y)$. Recall in theorem D.0.3, if $\Psi_{a,b}$ is a family of embeddings that has a singular limit satisfying the standing hypotheses D.0.3 then $\Psi_{a,b}$ admits an SRB measure for sufficiently small b . The standing hypotheses are satisfied by theorem 3.2.4, theorem 3.3.1 and 5.0.1. However, the family $\Psi_{a,b}$ fails to be an embedding, which is a vital condition in the construction of Wang and Young (2008) since it guarantees certain distortion bounds and subsequently guarantees the absolute continuity of a constructed invariant measure. It may be possible to construct such measures without needing the distortion bounds, which is a natural question for further research.

Proposition 5.0.1. Let $\Psi_{a,b}$ be the two-dimesional LSTM family as in (2.3). Let $F(x, y, a) = G(x + \frac{1}{2} \tanh(y), y, a)$. Then any critical point $\hat{y}(a)$ of the singular limit satisfies the nondegeneracy condition

$$\frac{\partial F(0, \hat{y}; a^*)}{\partial x} \neq 0. \tag{5.1}$$

Proof.

At a critical point $\hat{y}(a)$ we have

$$\begin{aligned} 0 &= F_y(0, \hat{y}(a), a) \\ &= G_x\left(\frac{1}{2} \tanh(y), y, a\right) \frac{1}{2} \operatorname{sech}^2(y) + G_y\left(\frac{1}{2} \tanh(y), y, a\right) \end{aligned}$$

Since $G_y(x, y, a) = \sigma(ax) > 0$. This implies that at a critical point $\hat{y}$ we must have

$$0 = G_x \left(\frac{1}{2} \tanh(\hat{y}(a)), \hat{y}(a), a \right) \frac{1}{2} \operatorname{sech}^2(\hat{y}(a)) + G_y \left(\frac{1}{2} \tanh(\hat{y}(a)), \hat{y}(a), a \right)$$

and so

$$\begin{aligned} -G_x \left(\frac{1}{2} \tanh(\hat{y}(a)), \hat{y}(a), a \right) \frac{1}{2} \operatorname{sech}^2(\hat{y}(a)) &= G_y \left(\frac{1}{2} \tanh(\hat{y}(a)), \hat{y}(a), a \right) > 0 \\ -F_x(0, \hat{y}(a), a) \frac{1}{2} \operatorname{sech}^2(\hat{y}(a)) &= G_y \left(\frac{1}{2} \tanh(\hat{y}(a)), \hat{y}(a), a \right) > 0 \end{aligned}$$

and since $\operatorname{sech}^2(y) > 0$ then $F_x(0, \hat{y}(a), a) \neq 0$ at critical points $\hat{y}$. $\square$

We now take some time to discuss the significance of this conjecture. The above addresses the attempts to satisfy the conditions of the following result from [Wang and Young \(2008\)](#).

Theorem 5.0.1. For an admissible family T_a as in [D.0.3](#), we assume that $\|T_a - F_a^\sharp\|_{C^3}$ is sufficiently small depending on $\{F_a\}$. Then there is a positive measure set $\Delta \subset [a_1, a_2]$ such that for all $a \in \Delta$, T_a admits an SRB measure.

In the case of transformations T_a that are not volume preserving, attractors are necessarily Lebesgue measure zero, which eliminates the possibility of an acim. The next best thing is an SRB measure which has a measure whose conditional measures on unstable manifolds are absolutely continuous. An equivalent condition is the existence of an open set U for which the conclusion of the Birkhoff ergodic theorem holds. In particular, this would imply observability of the positive Lyapunov exponents for points in U . This is an analogous result to the results proven above since it implies a significance to the chaotic behavior in terms of Lebesgue measure. For more information of SRB measures see [Young \(2002\)](#).

Chapter 6

Discussion

In this chapter, we conclude the paper with a discussion of the possible implications of the above results. This discussion can be summarized with the following two points. First, when selecting the LSTM model for a supervised learning task, it should be considered that LSTM can exhibit chaos and whether or not chaos is a desired property of the model chosen. Second, we hypothesize that chaos in the LSTM model may strain training and inference in a way that is related to the development of ResNets. Lastly, we mention some questions for further research.

As was discussed in section 1, the choice of the family of functions one chooses for a supervised learning task is very impactful. The results in this paper suggest that the possible chaotic behavior of LSTM should be considered. For example, if the data one wishes to model is inherently chaotic, this could be advantageous, although this would need to be explored in further detail. If chaos is not desired, then one would have to devise a way to avoid the chaotic regime or check after training that the optimal weights do not correspond to a chaotic LSTM.

We also hypothesize that this chaotic nature may create inefficiencies in inference and training. To see this, we first recall some of the issues remedied by ResNets. A standard feed forward network $\sigma(Wx)$ is a nonlinear function composed of a nonlinear function and a linear function. Thus it is impossible to find W such that $\sigma(Wx) = Id$. This becomes a problem in the following situation. Suppose you have a deep network with N layers and suppose that the first $N/2$ layers model the data perfectly. Then the last $N/2$ must approximate the identity function, which is difficult to do and will ultimately make training more difficult. We believe an inefficiency like this could exist in chaotic LSTM models. For

a given input sequence x_t , there is a desired output sequence h_t . In the chaotic regime, the “forcing” of x_t will have to overcome the baseline chaotic dynamics in order to obtain the correct output sequence. For more information on ResNets see [He et al. \(2020\)](#).

Finally, we conclude with some open questions for further research. The most immediate further research is the confirmation of the conjecture proposed in chapter 5 as well as similar results for higher dimensional LSTM models. The implications of these more general results would be that chaotic parameters make a positive measure set with respect to Lebesgue measure on the whole parameter space, meaning a nonzero probability of observing such chaos. This would truly mean that this chaos is non-negligable and practitioners must address it. If these results were to be proven false, then chaos would happen in a codimension 1 subspace which could be of no consequence in practice. Second, we should also ask what the geometry and topology of the chaotic regime are in the one dimensional case and in general. In the work of [Wang and Young \(2006\)](#), the “good parameters” are found to be a fat Cantor set and thus the chaotic parameters for the one dimensional LSTM also form a fat Cantor set. Does this persist in the higher dimensions and if so can we use this information to aid in training or inference? Lastly, more work should be done on the numerical and theoretical analysis of the forced case. While section 2.3.3 shows that the forced case can still be chaotic, one should still ask if we could characterize the input sequences which negate the unforced chaos.

Appendix A

Interval Analysis

Interval analysis is a mathematical technique for computing rounding, measurement and computational errors in mathematical computation and is an essential technique for computer assisted proofs in dynamical systems. Prominent examples of this are [Eckmann and Wittwer \(1987\)](#) and [Lanford III \(1982\)](#) in proving the Feigenbaum conjectures, [Tucker \(1999\)](#) in proving the existence of the Lorenz attractor and [Mischaikow et al. \(2001\)](#) for methods based on the Conley index. This section covers some of the fundamentals of interval analysis and computational methods we use in the above. We start by defining the interval number system and basic operations. Then we discuss interval functions and sequences of intervals. Lastly, we introduce Interval Newton's Method which we use to find the location of critical points. Much of the material below is based on [Moore et al. \(2009\)](#).

A.1 The Interval Number System

Let $\mathbb{IR}$ denote the set of all closed intervals of $\mathbb{R}$. Let $[x] \in \mathbb{IR}$ denote arbitrary interval of the form $[\underline{x}, \bar{x}]$. Two intervals $[x], [y]$ are equal if $\underline{x} = \underline{y}$ and $\bar{x} = \bar{y}$. If $[x]$ is such that $\underline{x} = \bar{x}$ then we call it *degenerate*. Later, we will see that $\mathbb{R}$ is embedded in $\mathbb{IR}$ as the set of degenerate intervals.

We can define the following operations on $\mathbb{IR}$ which are natural extensions of operations on $\mathbb{R}$ or set operations. Let the union and intersection of two intervals be as defined set-theoretically (note the union may not be an interval). Define the *interval hull*, *addition*, *subtraction*, *multiplication*, *inversion* and *division* as

$$[x] \sqcup [y] = [\min\{\underline{x}, \underline{y}\}, \max\{\bar{x}, \bar{y}\}]$$

$$[x] + [y] = [\underline{x} + \underline{y}, \bar{x} + \bar{y}]$$

$$[x] - [y] = [\underline{x} - \bar{y}, \bar{x} - \underline{y}]$$

$$[x] \cdot [y] = [\min S, \max S]$$

$$S = \{\underline{x}\underline{y}, \underline{x}\bar{y}, \bar{x}\underline{y}, \bar{x}\bar{y}\}$$

$$1/[x] = [1/\bar{x}, 1/\underline{x}]$$

$$\text{where } 0 \notin [x]$$

$$[x]/[y] = [x] \cdot (1/[y])$$

In addition, define the *width*, *absolute value* and *midpoint* as

$$w([x]) = \bar{x} - \underline{x}$$

$$|[x]| = \max\{|\underline{x}|, |\bar{x}|\}$$

$$m([x]) = \frac{1}{2}(\underline{x} + \bar{x})$$

We can define a metric on $\mathbb{IR}$ by

$$d([x], [y]) = \max\{|\underline{x} - \underline{y}|, |\bar{x} - \bar{y}|\}.$$

Under this metric, $\mathbb{R}$ is isometrically embedded into $\mathbb{IR}$ via the mapping $x \mapsto [x] = [x, x]$ since for degenerate intervals $[x], [y]$

$$\begin{aligned} d([x], [y]) &= d([x, x], [y, y]) \\ &= \max\{|\underline{x} - \underline{y}|, |\bar{x} - \bar{y}|\} \\ &= \max\{|x - y|, |x - y|\} \\ &= |x - y|. \end{aligned}$$

A.2 Interval Functions and Sequences

In this section, we study mappings $F : \mathbb{IR} \rightarrow \mathbb{IR}$ and sequences in $\mathbb{IR}$.

Given a mapping $g : \mathbb{R} \rightarrow \mathbb{R}$ (or $g : U \subset \mathbb{R} \rightarrow \mathbb{R}$) we can define a mapping $\bar{g} : \mathbb{IR} \rightarrow \mathbb{IR}$ called the *united extension*

$$\bar{g}([x]) = \bigcup_{a \in [x]} \{g(a)\}$$

Example A.2.1. A useful example of extending a function $g : \mathbb{R} \rightarrow \mathbb{R}$ to be defined on $\mathbb{IR}$ is the case of monotone functions. If g is monotone increasing, then we can define its extension

$$g([x]) = [g(\underline{x}), g(\bar{x})].$$

Definition A.2.1. We say that $F : \mathbb{IR} \rightarrow \mathbb{IR}$ is an interval extension of $f : \mathbb{R} \rightarrow \mathbb{R}$ if for all degenerate intervals we have

$$F([x, x]) = f(x).$$

Lemma A.2.1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ such that f is monotonically increasing on an interval $[x] = [\underline{x}, \bar{x}] \subset \mathbb{R}$. Let $F : \mathbb{IR} \rightarrow \mathbb{IR}$ be an interval function such that $f(x) \in F([x, x])$ for all $x \in \mathbb{R}$. Then we can construct a lower and upper bound of the image

$$[\overline{F(\underline{x})}, \underline{F(\bar{x})}] \subset f([x]) \subset [\underline{F(\underline{x})}, \overline{F(\bar{x})}].$$

Proof.

First note that $f(\underline{x}) \in F([\underline{x}, \underline{x}])$ and $f(\bar{x}) \in F([\bar{x}, \bar{x}])$. Then the following inequality holds

$$\begin{aligned} \underline{F([\underline{x}, \underline{x}])} &\leq f(\underline{x}) \leq \overline{F([\underline{x}, \underline{x}])} \\ \underline{F([\bar{x}, \bar{x}])} &\leq f(\bar{x}) \leq \overline{F([\bar{x}, \bar{x}])}. \end{aligned}$$

First, it is clear that $f([x]) \subset [\underline{F(\underline{x})}, \overline{F(\bar{x})}]$ since

$$\underline{F([\underline{x}, \underline{x}])} \leq f(\underline{x}) \leq f(\bar{x}) \leq \overline{F([\bar{x}, \bar{x}])}.$$

If $\overline{F(\underline{x}, \underline{x})} \leq \underline{F(\overline{x}, \overline{x})}$ then we have

$$f(\underline{x}) \leq \overline{F(\underline{x}, \underline{x})} \leq \underline{F(\overline{x}, \overline{x})} \leq f(\overline{x})$$

which proves the containment. Otherwise, the interval $[\overline{F(\underline{x})}, \underline{F(\overline{x})}]$ is empty, which also proves the containment true. $\square$

A.3 Interval Newton's Method

In this section we discuss Interval Newton's Method in one dimension. Let f be a continuously differentiable function on $[a, b]$. By the mean value theorem we have

$$f(x) = f(y) + f'(s)(x - y)$$

for some s between x and y . A zero x of f would satisfy

$$f(y) + f'(s)(x - y) = 0$$

for any y (note that s is thought of as a function of y). If we pick $y = m([a, b]) = (a + b)/2$ and we can write

$$x = y - f(y)/f'(s).$$

Applying this analysis to an inclusion monotonic interval extension $F'(x)$ of $f'(x)$, we get the iterative method

$$[x]^{(k+1)} = [x]^{(k)} \cap N([x]^{(k)}) \quad N([x]) = m([x]) - f(m([x]))/F'([x]). \quad (\text{A.1})$$

Theorem A.3.1. If an interval $[x]^{(0)}$ contains a zero x of $f(x)$, then so does $[x]^{(k)}$ for all k defined by (A.1). Furthermore, the intervals $[x]^{(k)}$ form a nested sequence converging to x if $0 \neq F'([x]^{(0)})$.

A.4 Machine Interval Arithmetic

In this section, we discuss the relationship between theoretical interval arithmetic and machine interval arithmetic. This is a necessary step in using interval arithmetic for computer-assisted proofs. Note that the code used for this paper is based on PyInterval which is built on CRLibm and follows IEEE-754 standards for correctly rounding elementary functions are machine interval arithmetic.

Recall that the set of floating point numbers are numbers of the form

$$x = m \cdot b^e$$

where m is called the *mantissa*, b is the *base* and e is the *exponent*. Numbers are normally represented in base $b = 2$ where $1/2 \leq m \leq 1$ and e is a bounded integer such that $e_{min} \leq e \leq e_{max}$. The set of *machine numbers* is denoted $\mathbb{R}_M$ and contains all numbers of the above form and their negatives.

A map $fl : \mathbb{R} \rightarrow \mathbb{R}_M$ is called an *optimal rounding* if

- (a) $x \leq y$ implies $fl(x) \leq fl(y)$ (monotonicity)
- (b) $fl(x) = x$ for all $x \in \mathbb{R}_M$.

A rounding is said to be directed downward (upward) if $fl(x) \leq x$ ($fl(x) \geq x$) for all $x \in \mathbb{R}$. For a downward directed rounding $\downarrow : \mathbb{R} \rightarrow \mathbb{R}_M$ we can define an upward directed rounding $\uparrow : \mathbb{R} \rightarrow \mathbb{R}_M$ by

$$\uparrow(x) := -(\downarrow(-x))$$

For any algorithm (sequence of elementary operations $+$, $-$, $\cdot$, $\div$) on real numbers, we can apply the same algorithm to machine numbers by replacing $x * y$ with $fl(x * y)$ for $*$ $\in \{+, -, \cdot, \div\}$.

The IEEE has four standard rounding schemes; round to nearest, round towards zero, round toward $+\infty$ and round towards $-\infty$. Using the appropriate rounding schemes, we can compute upper and lower bounds for the real output of the algorithm.

Example A.4.1. An important standard function to implement for machine numbers is the exponential function e^x . The implementation relies on the fact that for $x = E \ln(2) + z$ where $E \in \mathbb{Z}$ and $|z| \leq \ln(2)$

$$e^x = e^{E \ln(2) + z} = 2^E \cdot e^z.$$

This reduces the task of computing e^x to computing e^z where z falls into a small range. In fact, the range for z can be reduced even further, see [Daramy-Loirat et al. \(2006\)](#). The value of e^z is approximated by taylor polynomial and this approximation can be controlled to be a lower or upper bound. The exact implementation is beyond the scope of this paper. However, we can conclude that we can construct function $\exp_ru, \exp_rd : \mathbb{R}_M \rightarrow \mathbb{R}_M$ such that

$$\exp_rd(x) \leq e^x \leq \exp_ru(x).$$

One can use the same manner as above to implement machine intervals from real intervals. Let $\mathbb{IR}_M$ denote all closed intervals with endpoints in $\mathbb{R}_M$. Then we can define a rounding $\Downarrow : \mathbb{IR} \rightarrow \mathbb{IR}_M$ by

$$\Downarrow([x]) := [\downarrow(\underline{x}), \uparrow(\overline{x})].$$

This mapping has the properties

- (a) $[x] \subset [y]$ implies $\Downarrow([x]) \subset \Downarrow([y])$ for all $[x], [y] \in \mathbb{IR}$
- (b) $[x] \subset \Downarrow([x])$ for all $[x] \in \mathbb{IR}$

With these above properties, one can prove the following:

Theorem A.4.1. For $*$ $\in \{+, -, \cdot, \div\}$ and $[x], [y] \in \mathbb{IR}_M$ we have

$$[x] * [y] \subset \Downarrow([x] * [y]).$$

Thus, any algorithm (sequence of elementary operations) on real intervals can be implemented using machine intervals in the above manner and we will maintain the desired containment properties. Other elementary functions can also be implemented for machine intervals.

Example A.4.2. Recall the functions $\text{exp_ru}, \text{exp_rd} : \mathbb{R}_M \rightarrow \mathbb{R}_M$ from the previous example. We can use these function to implement a machine interval function $\text{exp} : \mathbb{IR}_M \rightarrow \mathbb{IR}_M$ by

$$\text{exp}([x]) = [\text{exp_rd}(\underline{x}), \text{exp_ru}(\overline{x})].$$

Note that this idea can be used to implement any monotonic function.

Appendix B

Formulae for LSTM

In this section, we list the formulae used for Ψ and its derivatives for the implementation of the rigorous numerics via interval arithmetic.

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (\text{B.1})$$

$$\sigma'(x) = \sigma(x)(1 - \sigma(x)) \quad (\text{B.2})$$

$$\sigma''(x) = \sigma'(x) = 2\sigma(x)\sigma'(x) \quad (\text{B.3})$$

$$\tanh(x) = \frac{e^{2x} - 1}{e^{2x} + 1} \quad (\text{B.4})$$

$$\text{sech}(x) = \frac{2}{e^x + e^{-x}} \quad (\text{B.5})$$

$$\tanh'(x) = \text{sech}^2(x) \quad (\text{B.6})$$

$$\tanh''(x) = -2 \tanh(x) \text{sech}^2(x) \quad (\text{B.7})$$

$$\tanh'''(x) = -4 \tanh^2(x) \text{sech}^2(x) - 2 \text{sech}^4(x) \quad (\text{B.8})$$

$$F(x) = \frac{1}{2} \tanh(x) \quad (\text{B.9})$$

$$F'(x) = \frac{1}{2} \text{sech}^2(x) \quad (\text{B.10})$$

$$G(x, y, a) = \sigma(ax)y + \sigma(k_1x) \tanh(k_2x) \quad (\text{B.11})$$

$$\Psi_a(y) = G(F(y), y, a) \quad (\text{B.12})$$

$$\frac{\partial}{\partial a} \Psi_a(y) = \frac{1}{2} \tanh(y) \sigma'(a \frac{1}{2} \tanh(y)) y \quad (\text{B.13})$$

$$\Psi'_a(y) = G_x(F(y), y, a) F'(y) + G_y(F(y), y, a) \quad (\text{B.14})$$

$$\Psi''_a(y) = (G_{xx}(F(y), y, a) F'(y) + 2G_{xy}(F(y), y, a)) F'(y) \quad (\text{B.15})$$

$$\begin{aligned}
& + G_x(F(y), y, a)F''(y) \\
\Psi_a'''(y) & = G_{xxx}(F(y), y, a)(F'(y))^3 + 3G_{xxy}(F(y), y, a)(F'(y))^2
\end{aligned} \tag{B.16}$$

$$\begin{aligned}
& + 3G_{xx}(F(y), y, a)F''(y)F'(y) + 3G_{xy}(F(y), y, a)F''(y) \\
& + G_x(F(y), y, a)F'''(y) \\
S(\Psi_a)(y) & = \frac{\Psi_a'''(y)}{\Psi_a'(y)} - \frac{3}{2} \left(\frac{\Psi_a''(y)}{\Psi_a'(y)} \right)^2
\end{aligned} \tag{B.17}$$

$$G_x(x, y, a) = a\sigma'(ax)y + k_1\sigma'(k_1x)\tanh(k_2x) + k_2\sigma(k_1x)\operatorname{sech}^2(k_2x) \tag{B.18}$$

$$G_y(x, y, a) = \sigma(ax) \tag{B.19}$$

$$\begin{aligned}
G_{xx}(x, y, a) & = a^2\sigma''(ax)y + k_1^2\sigma''(k_1x)\tanh(k_2x) \\
& + 2k_1k_2\sigma'(k_1x)\operatorname{sech}^2(k_2x) + k_2^2\sigma(k_1x)??
\end{aligned} \tag{B.20}$$

$$G_{xy}(x, y, a) = a\sigma'(ax) \tag{B.21}$$

$$G_{xxx}(x, y, a) = a^3\sigma'''(ax)y + k_1^3\sigma'''(k_1x)\tanh(k_2x) \tag{B.22}$$

$$+ 3k_1^2k_2\sigma''(k_1x)\tanh'(k_2x) \tag{B.23}$$

$$+ 3k_1k_2^2\sigma'(k_1x)\tanh''(k_2x) + k_2^3\sigma(k_1x)\tanh'''(k_2x) \tag{B.24}$$

$$G_{xxy}(x, y, a) = a^2\sigma''(ax) \tag{B.25}$$

Appendix C

One Dimensional and Symbolic Dynamics

In this appendix we give some background on topological and symbolic dynamics. The material in this section is based on [Bruin \(2023\)](#).

C.1 One Dimensional Dynamics

Definition C.1.1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a C^3 map such that $f' \neq 0$. The *Schwarzian derivative* is defined as

$$Sf(x) = \frac{f'''(x)}{f'(x)} - \frac{3}{2} \left(\frac{f''(x)}{f'(x)} \right)^2$$

Proposition C.1.1. Let $f, g \in C^3(\mathbb{R})$ such that $Sf, Sg < 0$. Then $S(f \circ g) < 0$.

The important corollary to this is that for f with $Sf < 0$ we have $S(f^n) < 0$ for all n .

C.2 Symbolic Dynamics

In this section, we discuss a few results from symbolic dynamics.

For a finite or countable alphabet $\mathcal{A}$ we can consider the space of sequences $\Sigma = \mathcal{A}^{\mathbb{N}}$ with the Tychonoff topology. The map $\sigma : \Sigma \rightarrow \Sigma$ defined by $\sigma(x)_i = x_{i+1}$ is called the shift map. A pair (X, σ) is called a subshift if X is strongly shift-invariant (i.e. $\sigma(X) = X$) and X is topologically closed. Any finite sequence of letters that appears in some sequence in X is called a word. The set of all words is called the language and is denoted $\mathcal{L}(X)$.

Definition C.2.1. Let X be a compact Hausdorff space and let C be a finite open cover of X . For two covers C, D of X let $C \vee D$ be the minimal common refinement. Let $H(C)$

denote the minimal number of elements of C required to cover X . Then for a continuous map $f : X \rightarrow X$ define

$$H(f, C) = \lim_{n \rightarrow \infty} \frac{1}{n} H(C \vee f^{-1}C \vee \dots \vee f^{-n+1}C).$$

Then we define the *topological entropy* of f to be $h_{top}(f) = \sup_C H(f, C)$.

Definition C.2.2. A subshift is said to have *finite-type* if it consists of all sequences avoiding a finite set of words as subwords.

It can be shown that every subshift of finite type with finite alphabet can be represented by a transition graph ...

Definition C.2.3. A subshift is said to be *irreducible* if for every $u, v \in \mathcal{L}(X)$ there is a word $w \in \mathcal{L}(X)$ such that $uvw \in \mathcal{L}(X)$. For subshifts of finite type, this is equivalent to the transition matrix being irreducible (i.e. for each (i, j) entry there exists k such that $A^k(i, j) > 0$).

Lemma C.2.1. Let (X, f) be an irreducible subshift of finite type with transition matrix A . Then $h_{top}(f) = \log \lambda$ where λ is the largest eigenvalue of A .

Theorem C.2.1. Let (X, f) and (Y, g) be conjugate dynamical systems. Then $h_{top}(f) = h_{top}(g)$.

Appendix D

On Theory of Rank One Chaos

This section reviews some results from the rank one theory and nonuniformly expanding 1D maps, primarily from [Wang and Young \(2006\)](#) and [Wang and Young \(2008\)](#). More information can be found in [Wang and Young \(2001\)](#) and [Wang and Young \(2002\)](#)

Definition D.0.1 (Misiurewicz Map). Let $f \in C^2(I, I)$ where I is a closed interval or S^1 . Let C denote the set of critical points of f and $C_\delta = \{x \in I : \text{dist}(x, C) < \delta\}$ for some $\delta > 0$. We call f a *Misiurewicz map* if there exists some $\delta_0 > 0$ such that

(a) Outside of C_{δ_0} : there exists $\lambda_0 > 1$, $M_0 \in \mathbb{N}$ and $0 < c_0 \leq 1$ such that

(i) for all $n \geq M_0$, if $x, f(x), \dots, f^{n-1}(x) \notin C_{\delta_0}$, then $|(f^n)'(x)| \geq e^{\lambda_0 n}$;

(ii) if $x, f(x), \dots, f^{n-1}(x) \notin C_{\delta_0}$ and $f^n(x) \in C_{\delta_0}$, any n , then $|(f^n)'(x)| \geq c_0 e^{\lambda_0 n}$.

(b) Inside of C_{δ_0} :

(i) $f''(x) \neq 0$ for all $x \in C_{\delta_0}$;

(ii) for all $\hat{x} \in C$ and $n > 0$, $d(f^n(\hat{x}), C) \geq \delta_0$;

(iii) for all $x \in C_{\delta_0} \setminus C$, there exists $p_0(x) > 0$ such that $f^j(x) \notin C_{\delta_0}$ for all $j < p_0(x)$ and $|(f^{p_0(x)})'(x)| \geq c_0^{-1} e^{\frac{1}{3}\lambda_0 p_0(x)}$.

We denote the set of all Misiurewicz maps (on a given domain) by $\mathcal{M}$.

Definition D.0.2 (Good Maps). For $\lambda, \alpha, \varepsilon > 0$, $f_0 \in \mathcal{M}$ and $f \in C^2(I, I)$ we say $f \in \mathcal{G}(f_0; \lambda, \alpha, \varepsilon)$ if $\|f_0 - f\|_{C^2} < \varepsilon$ and

(a) $d(f^n(\hat{x}), C) > \min\{\frac{1}{2}\delta_0, e^{-\alpha n}\}$

$$(b) \quad |(f^n)'(f(\hat{x}))| \geq c_1 e^{\lambda n}$$

where $\hat{x} \in C = C(f)$, $n > 0$ and $c_1 = \min\{c_0, c'_0, c''_0\}$ as in Wang and Young (2006). Then $\mathcal{G}$ is the union of all $\mathcal{G}(f_0; \lambda, \alpha, \varepsilon)$ that satisfy condition (P1)-(P3) in Wang and Young (2006).

Theorem D.0.1. Good maps admit absolutely continuous invariant measures.

Theorem D.0.2. For every C^2 family $\{f_a\}$ satisfying the parameter transversality condition at $f_{a^*} \in \mathcal{M}$, the set $\{a : f_a \in \mathcal{G}\}$ has positive Lebesgue measure.

Definition D.0.3 (Standing Hypotheses). We consider embedding $T_a : X \rightarrow X$, $a \in [a_1, a_2]$ where $\|T_a - F^\sharp\|_{C^3}$ is small for some F_a satisfying the following conditions:

- (a) (Misiurewicz Map) There exists $a^* \in [a_1, a_2]$ such that $f_{a^*} \in \mathcal{M}$.
- (b) (Parameter Transversality) For every $\hat{x} \in C(f_{a^*})$ and $q = f_{a^*}(\hat{x})$.

$$\frac{d}{da} f_a(\hat{x}(a)) \neq \frac{d}{da} q(a) \text{ at } a = a^*. \quad (\text{D.1})$$

- (c) (Nondegeneracy) For every $\hat{x} \in C$, there exists $j \leq m - 1$ such that

$$\frac{\partial F(\hat{x}, 0; a^*)}{\partial y^j} \neq 0. \quad (\text{D.2})$$

The family T_a is said to satisfying the *standing hypotheses*.

Theorem D.0.3. In addition to the standing hypothese above, we assume that $\|T_a - F_a^\sharp\|_{C^3}$ is sufficiently small depending on $\{F_a\}$. Then there is a positive measure set $\Delta \subset [a_1, a_2]$ such that for all $a \in \Delta$, T_a admits an SRB measure.

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